

Contents**1 Misc 2**

1.1	Default Code	2
1.2	Run.py	2
1.3	Diff	2
1.4	Hash Command	2
1.5	Run.sh	2
1.6	Run.bat	2
1.7	Custom Set PQ Sort	2
1.8	Dynamic Bitset	2
1.9	Enumerate Subset	2
1.10	Fast Input	2
1.11	OEIS	2
1.12	OEIS	2
1.13	Pragma	3
1.14	Python	3
1.15	Xor Basis	3
1.16	disable ASLR	3
1.17	hash windows	3
1.18	random int	3

2 Convolution 3

2.1	FFT	3
2.2	FFT any mod	4
2.3	FWT	4
2.4	Min Convolution Concave Concave	4
2.5	NTT mod 998244353	4
2.6	PolyInv	4
2.7	PolySqrt	5

3 Data-Structure 5

3.1	BIT	5
3.2	Disjoint Set Persistent	5
3.3	PBDS GP Hash Table	5

3.4 PBDS Order Set 6

3.5	Segment Tree Add Set	6
3.6	Segment Tree Li Chao Line	6
3.7	Segment Tree Li Chao Segment	7
3.8	Segment Tree Persistent	7
3.9	Segment Tree Zkw	7
3.10	Sparse Table	8
3.11	Treap2	8
3.12	Trie	8

4 Dynamic-Programming 9

4.1	Digit DP	9
4.2	Integer Partition	9
4.3	Knapsack On Tree	9
4.4	SOS DP	9

5 Geometry 9

5.1	misc	9
5.2	convex hull	9
5.3	SVG writer	10
5.4	struct point	10
5.5	point / basic operation	10
5.6	point / operators	10
5.7	point / banana	10
5.8	point / rayHitSeg	10
5.9	point / else	10
5.10	struct line	10
5.11	line / else	10
5.12	struct polygon	10
5.13	polygon / make convex hull	10
5.14	polygon / in polygon	11
5.15	polygon / in convex	11
5.16	polygon / cycle search	11
5.17	polygon / line cut convex	11
5.18	polygon / convex line intersect	11

5.19 polygon / segment pass con-

vex interior	11
5.20 polygon / convex tangent point	11

5.21 polygon / halfplane intersection 11

5.22 polygon / minkowski sum	12
5.23 struct Cir	12
5.24 Cir / Cir intersect	12
5.25 Cir / point tangent	12
5.26 Cir / circle tangent	12
5.27 Cir / minimum enclosing circle	12
5.28 Geometry Struct 3D	12
5.29 Pick's Theorem	13

6 Graph 13

6.1	SAT	13
6.2	Augment Path	13
6.3	C3C4	14
6.4	Dinic	14
6.5	Dominator Tree	14
6.6	EdgeBCC	15
6.7	EnumeratePlanarFace	15
6.8	Ford Fulkerson	15
6.9	HLD	16
6.10	Kosaraju SCC	16
6.11	Kuhn Munkres	16
6.12	LCA	17
6.13	MCMF	17
6.14	Tarjan SCC	18
6.15	Theorem	18
6.16	Tree Isomorphism	18
6.17	prufer reverse	18
6.18	圓方樹	18
6.19	最大權閉合圖	19

7 Math 19

7.1	Burnside's Lemma	19
7.2	CRT	19
7.3	Catalan Number	19
7.4	Floor Sum	19
7.5	Josephus Problem	19
7.6	Lagrange any x	19
7.7	Lagrange continuous x	19
7.8	Linear Mod Inverse	20
7.9	Lucas's Theorem	20
7.10	Matrix	20
7.11	Matrix 01	20
7.12	Matrix Tree Theorem	20
7.13	Miller Rabin	21
7.14	Pollard Rho	21
7.15	Stirling's formula	21
7.16	Theorem	21
7.17	josephus	21
7.18	二元一次方程式	21
7.19	數論分塊	21
7.20	最大質因數	21
7.21	歐拉公式	21
7.22	歐拉定理	22
7.23	錯排公式	22

8 String 22

8.1	AC automation	22
8.2	Enumerate Runs	22
8.3	Hash	22
8.4	KMP	22
8.5	Manacher	23
8.6	Min Rotation	23
8.7	Suffix Array	23
8.8	Suffix Automata	24
8.9	Wildcard Matching	24
8.10	Z Algorithm	24

1 Misc

1.1 Default Code [24a798]

```
1 #include <bits/stdc++.h>
2 using namespace std;
3 #define int long long
4 #define debug(a...) cerr << #a << " = ", dout(a)
5 void dout() { cerr << "\n"; }
6 template<typename A, typename... B>
7 void dout(A a, B... b) { cerr << a << ' ', dout(b...); }
8
9 void solve(){
10
11 }
12
13 signed main(){
14     ios::sync_with_stdio(0), cin.tie(0);
15
16     int t = 1;
17     while (t--){
18         solve();
19     }
20
21     return 0;
22 }
```

1.2 Run.py

```
1 from os import *
2 f = "pA"
3 while 1:
4     i = input("> ") or f; system("clear"); f = i
5     print(f"file = {f}")
6     if system(f"g++ {f}.cpp -std=c++17 -Wall -Wextra -Wshadow
7         -O2 -D LOCAL -g -fsanitize=undefined,address -o {f}
8         "):
9         print("CE"); continue
10    for x in sorted(listdir()):
11        if x.startswith(f) and x.endswith(".in"):
12            print(x); system(f"./{f} < {x}") and print("RE");
13            print()
```

1.3 Diff

```
1 set -e
2 g++ ac.cpp -o ac
3 g++ wa.cpp -o wa
4 for ((i=0;i++))
5 do
6     echo "$i"
7     python3 gen.py > input
8     ./ac < input > ac.out
9     ./wa < input > wa.out
10    diff ac.out wa.out || break
11 done
```

1.4 Hash Command

```
1 cat file.cpp | cpp -dD -P -fpreprocessed | tr -d "[:space:]"
2 | md5sum | cut -c-6
```

1.5 Run.sh

```
1 if [ ! -f $1 ] || [ $1.cpp -nt $1 ]; then
2     g++ $1.cpp -std=c++17 -Wall -Wextra -Wshadow -O2 -D LOCAL
3     -g -fsanitize=undefined,address -o $1 ||
4     { echo "Compile failed."; exit 1; }
5 fi
6
7 ./ $1
```

1.6 Run.bat

```
1 @echo off
2 for /f %i in ('dir /b /od %1.cpp %1.exe 2^>nul') do set L=%i
3 if not "%L%"=="%1.exe" (
4     g++ %1.cpp -std=c++17 -Wall -Wextra -Wshadow -O2 -D LOCAL ^
5     -g -o %1 || (echo Compile failed.& exit /b 1)
6 )
7 %L
```

1.7 Custom Set PQ Sort [d4df55]

```
1 // 所有自訂的結構體，務必檢查相等的 case，給所有元素一個排序
2 // 的依據
3 struct my_struct{
4     int val;
5     my_struct(int _val) : val(_val) {}
6 };
7
8 auto cmp = [](my_struct a, my_struct b) {
9     return a.val > b.val;
10 };
11
12 set<my_struct, decltype(cmp)> ss({1, 2, 3}, cmp);
13 priority_queue<my_struct, vector<my_struct>, decltype(cmp)>
14     pq(cmp, {1, 2, 3});
15 map<my_struct, my_struct, decltype(cmp)> mp({{1, 4}, {2, 5},
16     {3, 6}}, cmp);
```

1.8 Dynamic Bitset [c78aa8]

```
1 const int MAXN = 2e5 + 5;
2 template<int len = 1>
3 void solve(int n) {
4     if (n > len) {
5         solve<min(len*2, MAXN)>(n);
6         return;
7     }
8     bitset<len> a;
9 }
```

1.9 Enumerate Subset [a13e46]

```
1 // 時間複雜度  $O(3^n)$ 
2 // 枚舉每個 mask 的子集
3 for (int mask=0; mask<(1<n); mask++){
4     for (int s=mask; s>=0; s=(s-1)&m){
5         // s 是 mask 的子集
6         if (s==0) break;
7     }
8 }
```

1.10 Fast Input [6f8879]

```
1 // fast IO
2 // 6f8879
3 inline char readchar(){
4     static char buffer[BUFSIZ], *now = buffer + BUFSIZ, *
5     end = buffer + BUFSIZ;
6     if (now == end)
7     {
8         if (end < buffer + BUFSIZ)
9             return EOF;
10        end = (buffer + fread(buffer, 1, BUFSIZ, stdin));
11        now = buffer;
12    }
13    return *now++;
14 }
15 inline int nextint(){
16     int x = 0, c = readchar(), neg = false;
17     while(('0' > c || c > '9') && c!='-' && c!=EOF) c =
18         readchar();
19     if(c == '-') neg = true, c = readchar();
20     while('0' <= c && c <= '9') x = (x<<3) + (x<<1) + (c^'0')
21         , c = readchar();
22     if(neg) x = -x;
23     return x; // returns 0 if EOF
```

1.11 OEIS [25cf74]

```
1 // 若一個線性遞迴有 k 項，給他恰好 2*k 個項可以求出線性遞迴
2 // s = 1 1 3 7 17，則它會回傳 1 2
3 vector<int> BerlekampMassey(vector<int> s) {
4     if (s.empty()) return vector<int>();
5     int n = s.size(), L = 0, m = 0;
6     vector<int> C(n), B(n), T;
7     C[0] = B[0] = 1;
8
9     int b = 1;
10    for (int i=0; i<n; i++) {
11        ++m; int d = s[i]%MOD;
12        for (int j=1; j<L+1; j++) d = (d+C[j]*s[i-j])%MOD;
13        if (!d) continue;
14        T = C; int coef = d*qp(b, MOD-2, MOD)%MOD;
15        for (int j=m; j<n; j++) C[j] = (C[j]-coef*B[j-m])%
16            MOD;
17        if (2*L>i) continue;
18        L = i+1-L; B=T; b=d; m=0;
19    }
20    C.resize(L+1); C.erase(C.begin());
21    for (int &x : C) x = (MOD-x)%MOD;
22    reverse(C.begin(), C.end());
23    return C;
24 }
```

1.12 OEIS

```
1 from fractions import Fraction as F
2 def BM(s):
3     n = len(s); L=m=0; C=[F(0)]*n; B=[F(0)]*n; C[0]=B[0]=F(1)
4     ; b=F(1)
5     for i in range(n):
6         m+=1; d=sum(C[j]*s[i-j] for j in range(L+1))
7         if d==0: continue
8         T = C[:]; coef = F(d/b)
```

```

8         for j in range(m, n): C[j] -= coef*B[j-m]
9         if 2*L<=i: L, B, b, m = i+1-L, T, d, 0
10        if len(C)<L+1: C += [F(0)] * (L+1-len(C))
11        return [-x for x in C[1:L+1]]

```

1.13 Pragma [09d13e]

```

1 #pragma GCC optimize("O3,unroll-loops")
2 #pragma GCC target("avx,avx2,sse,sse2,sse3,sse4,popcnt")

```

1.14 Python

```

1 # Decimal
2 from decimal import *
3 getcontext().prec = 6
4
5 # system setting
6 sys.setrecursionlimit(100000)
7 sys.set_int_max_str_digits(10000)
8
9 # turtle
10 from turtle import *
11
12 N = 3000000010
13 setworldcoordinates(-N, -N, N, N)
14 hideturtle()
15 speed(100)
16
17 def draw_line(a, b, c, d):
18     teleport(a, b)
19     goto(c, d)
20
21 def write_dot(x, y, text, diff=1): # diff = 文字的偏移
22     teleport(x, y)
23     dot(5, "red")
24
25     teleport(x+N/100*diff, y+N/100*diff)
26     write(text, font=("Arial", 5, "bold"))
27
28 # usage
29 draw_line(*a[i], *(a[i-1]))
30 write_dot(*a[i], str(a[i]))
31 # 以自己左方 100 單位為圓心向前畫一個 90 度的弧 / 正五邊形
   (參數可以是負的)
32 circle(100, extent = 90)
33 circle(100, steps = 5)
34 left(90) # 左轉 90 度
35 fd(-100) # 倒退 100 單位
36 penup()
37 pendown()
38
39 tracer(0, 0) # IO 優化 · 必須在程式碼最後手動 update()
40 pos() # 回傳目前座標
41 heading() # 回傳目前方向：0 是右方、90 是上方
42 color("red") # 改變顏色 · 可以用文字或是 #000000 的格式
43
44 # OOP
45 class Point:
46     def __init__(self, x, y):
47         self.x = x
48         self.y = y
49
50     def __add__(self, o): # use dir(int) to know operator
       name

```

```

51         return Point(self.x+o.x, self.y+o.y)
52
53     @property
54     def distance(self):
55         return (self.x**2 + self.y**2)**(0.5)
56
57 a = Point(3, 4)
58 print(a.distance)
59
60 # Fraction
61 from fractions import Fraction
62 a = Fraction(Decimal(1.1))
63 a.numerator # 分子
64 a.denominator # 分母

```

1.15 Xor Basis [123f2d]

```

1 vector<int> basis;
2 void add_vector(int x){
3     for (auto v : basis){
4         x=min(x, x^v);
5     }
6     if (x) basis.push_back(x);
7 }
8
9 // 給一數字集合 S · 求能不能 XOR 出 x
10 bool check(int x){
11     for (auto v : basis){
12         x=min(x, x^v);
13     }
14     return (x == 0);
15 }
16
17 // 給一數字集合 S · 求能 XOR 出多少數字
18 // 答案等於 2^{basis 的大小}
19
20 // 給一數字集合 S · 求 XOR 出最大的數字
21 int get_max(){
22     int ans=0;
23     for (auto v : basis){
24         ans=max(ans, ans^v);
25     }
26     return ans;
27 }

```

1.16 disable ASLR [29a49f]

```

1 // Disable randomization of memory addresses
2 // setarch $(uname -m) -R /bin/bash
3
4 #include <sys/personality.h>
5 #include <bits/stdc++.h>
6 int main(int argc, char* argv[]){
7     personality(ADDR_NO_RANDOMIZE);
8     execv(argv[1], argv+1);
9 }

```

1.17 hash windows

```

1 def get_hash(path):
2     from subprocess import run, PIPE
3     from hashlib import md5
4

```

```

5     p = run(
6         ["cpp", "-dD", "-P", "-fpreprocessed", path],
7         stdout = PIPE,
8         stderr = PIPE,
9         text = True
10    )
11
12    if p.returncode != 0:
13        raise RuntimeError(p.stderr)
14
15    s = ''.join(p.stdout.split())
16    ret = md5(s.encode()).hexdigest()
17    return ret[:6]
18
19 print(get_hash("Suffix_Array.cpp"))

```

1.18 random int [9cc603]

```

1 mt19937 seed(chrono::steady_clock::now().time_since_epoch().
   count());
2 int rng(int l, int r){
3     return uniform_int_distribution<int>(l, r)(seed);
4 }

```

2 Convolution

2.1 FFT [16591a]

```

1 typedef complex<double> cd;
2
3 // 778272
4 void FFT(vector<cd> &a) {
5     int n = a.size(), L = 31-__builtin_clz(n);
6     vector<complex<long double>> R(2, 1);
7     vector<cd> rt(2, 1);
8     for (int k=2 ; k<n ; k*=2){
9         R.resize(n), rt.resize(n);
10        auto x = polar(1.0L, acos(-1.0L) / k);
11        for (int i=k ; i<2*k ; i++) rt[i] = R[i] = (i&1 ? R[i/2]*x : R[i/2]);
12    }
13
14    vector<int> rev(n);
15    for (int i=0 ; i<n ; i++) rev[i] = (rev[i/2] | (i&1)<<L)/2;
16
17    for (int i=0 ; i<n ; i++) if (i<rev[i]) swap(a[i], a[rev[i]]);
18
19    for (int k=1 ; k<n ; k*=2){
20        for (int i=0 ; i<n ; i+=2*k){
21            for (int j=0 ; j<k ; j++){
22                // cd z = rt[j+k] * a[i+j+k];
23                auto x = (double *)&rt[j+k], y = (double *)&a[i+j+k];
24                cd z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] + x[1]*y[0]);
25                a[i+j+k] = a[i+j]-z;
26                a[i+j] += z;
27            }
28        }
29    }
30
31 // 192528
32 vector<double> PolyMul(const vector<double> a, const vector<double> b){

```

```

32 if (a.empty() || b.empty()) return {};
33 vector<double> res(a.size()+b.size()-1);
34 int L = 32 - __builtin_clz(res.size()), n = 1<<L;
35 vector<cd> in(n), out(n);
36 copy(a.begin(), a.end(), in.begin());
37 for (int i=0 ; i<b.size() ; i++) in[i].imag(b[i]);
38 FFT(in);
39 for (cd& x : in) x *= x;
40 for (int i=0 ; i<n ; i++) out[i] = in[-i & (n - 1)] -
    conj(in[i]);
41 FFT(out);
42 for (int i=0 ; i<res.size() ; i++) res[i] = imag(out[i])
    / (4 * n);
43 return res;
44 }

```

2.2 FFT any mod [412000]

```

1 // n=524288, ~400ms
2 const int MOD = 998244353;
3 typedef complex<double> cd;
4
5 // bb6d94
6 void FFT(vector<cd> &a) {
7     int n = a.size(), L = 31-__builtin_clz(n);
8     vector<complex<long double>> R(2, 1);
9     vector<cd> rt(2, 1);
10    for (int k=2 ; k<n ; k*=2){
11        R.resize(n);
12        rt.resize(n);
13        auto x = polar(1.0L, acos(-1.0L) / k);
14        for (int i=k ; i<2*k ; i++){
15            rt[i] = R[i] = (i&1 ? R[i/2]*x : R[i/2]);
16        }
17    }
18
19    vector<int> rev(n);
20    for (int i=0 ; i<n ; i++) rev[i] = (rev[i/2] | (i&1)<<L)
        /2;
21    for (int i=0 ; i<n ; i++) if (i<rev[i]) swap(a[i], a[rev[
        i]]);
22    for (int k=1 ; k<n ; k*=2){
23        for (int i=0 ; i<n ; i+=2*k){
24            for (int j=0 ; j<k ; j++){
25                auto x = (double *)&rt[j+k];
26                auto y = (double *)&a[i+j+k];
27                cd z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] + x[1]*
                    y[0]);
28                a[i+j+k] = a[i+j]-z;
29                a[i+j] += z;
30            }
31        }
32    }
33 }
34
35 // c3dcf6
36 vector<int> PolyMul(vector<int> a, vector<int> b){
37     if (a.empty() || b.empty()) return {};
38     vector<int> res(a.size()+b.size()-1);
39     int B = 32-__builtin_clz(res.size()), n = (1<<B), cut = (
        int)(sqrt(MOD));
40     vector<cd> L(n), R(n), outs(n), outl(n);
41
42     for (int i=0 ; i<a.size() ; i++) L[i] = cd((int) a[i]/cut
        , (int)a[i]%cut);

```

```

43     for (int i=0 ; i<b.size() ; i++) R[i] = cd((int) b[i]/cut
        , (int)b[i]%cut);
44     FFT(L), FFT(R);
45     for (int i=0 ; i<n ; i++){
46         int j = -i&(n-1);
47         outl[j] = (L[i]+conj(L[j])) * R[i]/(2.0*n);
48         outs[j] = (L[i]-conj(L[j])) * R[i]/(2.0*n)/1i;
49     }
50     FFT(outl), FFT(outs);
51     for (int i=0 ; i<res.size() ; i++){
52         int av = (int)(real(outl[i])+0.5), cv = (int)(imag(
            outs[i])+0.5);
53         int bv = (int)(imag(outl[i])+0.5) + (int)(real(outs[i
            ])+0.5);
54         res[i] = ((av%MOD*cut+bv) % MOD*cut+cv) % MOD;
55     }
56     return res;
57 }

```

2.3 FWT [e50788]

```

1 // 已經把 mint 刪掉 · 需要增加註解
2 vector<int> xor_convolution(vector<int> a, vector<int> b, int
    k) {
3     if (k==0) return vector<int>{a[0]*b[0]};
4     vector<int> aa(1<<(k-1)), bb(1<<(k-1));
5     for (int i=0 ; i<(1<<(k-1)) ; i++){
6         aa[i] = a[i] + a[i + (1<<(k-1))];
7         bb[i] = b[i] + b[i + (1<<(k-1))];
8     }
9     vector<int> X = xor_convolution(aa, bb, k-1);
10    for (int i=0 ; i<(1<<(k-1)) ; i++){
11        aa[i] = a[i] - a[i + (1<<(k-1))];
12        bb[i] = b[i] - b[i + (1<<(k-1))];
13    }
14    vector<int> Y = xor_convolution(aa, bb, k-1);
15    vector<int> c(1<<k);
16    for (int i=0 ; i<(1<<(k-1)) ; i++){
17        c[i] = (X[i] + Y[i]) / 2;
18        c[i + (1<<(k-1))] = (X[i] - Y[i]) / 2;
19    }
20    return c;
21 };

```

2.4 Min Convolution Concave Concave [ffb28d]

```

1 // 需要增加註解
2 // min convolution
3 vector<int> mkk(vector<int> a, vector<int> b) {
4     vector<int> slope;
5     FOR (i, 1, ssize(a) - 1) slope.pb(a[i] - a[i - 1]);
6     FOR (i, 1, ssize(b) - 1) slope.pb(b[i] - b[i - 1]);
7     sort(all(slope));
8     slope.insert(begin(slope), a[0] + b[0]);
9     partial_sum(all(slope), begin(slope));
10    return slope;
11 }

```

2.5 NTT mod 998244353 [976c8d]

```

1 const int MOD = (119 << 23) + 1, ROOT = 62; // = 998244353
2 // For p < 2^30 there is also e.g. 5 << 25, 7 << 26, 479 <<
    21
3 // and 483 << 21 (same root). The last two are > 10^9.

```

```

4 // 7e041b
5 vector<int> rt(2, 1);
6 void NTT_init(int n){
7     static bool lastInit = 0;
8     if (lastInit<n){
9         lastInit = n;
10        for (int k=2, s=2 ; k<n ; k*=2, s++){
11            rt.resize(n);
12            int z[] = {1, qp(ROOT, MOD>>s, MOD)};
13            for (int i=k ; i<2*k ; i++) rt[i] = rt[i/2]*z[i
                &1]%MOD;
14        }
15    }
16 }
17
18 // fec4b8
19 void NTT(vector<int> &a, int isInverse = false) {
20     int n = a.size();
21     int L = 31-__builtin_clz(n);
22     NTT_init(n);
23
24     vector<int> rev(n);
25     for (int i=0 ; i<n ; i++) rev[i] = (rev[i/2] | (i&1)<<L)/2;
26     for (int i=0 ; i<n ; i++){
27         if (i<rev[i]) swap(a[i], a[rev[i]]);
28     }
29
30     for (int k=1 ; k<n ; k*=2){
31         for (int i=0 ; i<n ; i+=2*k){
32             for (int j=0 ; j<k ; j++){
33                 int z = rt[j+k]*a[i+j+k]%MOD, &ai = a[i+j];
34                 a[i+j+k] = ai-z+(z*ai ? MOD : 0);
35                 ai += (ai+z)>MOD ? z-MOD : z;
36             }
37         }
38     }
39
40     if (isInverse) {
41         reverse(a.begin()+1, a.end());
42         int n_inv = qp(n, MOD-2, MOD);
43         for (int i=0 ; i<n ; i++) a[i] = a[i]*n_inv%MOD;
44     }
45 }
46
47 // 771576
48 vector<int> polyMul(vector<int> a, vector<int> b){
49     if (a.empty() || b.empty()) return {};
50     int s = a.size()+b.size()-1, B = 32-__builtin_clz(s), n =
        1<<B;
51     vector<int> L(a), R(b), out(n);
52     L.resize(n), R.resize(n);
53     NTT(L), NTT(R);
54     for (int i=0 ; i<n ; i++) out[i] = L[i]*R[i]%MOD;
55     NTT(out, true);
56     out.resize(s);
57     return out;
58 }
59
60 // 771576
61 vector<int> polyInv(vector<int> a){
62     int n = a.size();
63     vector<int> ret(1, qp(a[0], MOD-2, MOD));
64 }

```

2.6 PolyInv [b388fb]

```

1 vector<int> polyInv(vector<int> a){
2     int n = a.size();
3     vector<int> ret(1, qp(a[0], MOD-2, MOD));
4 }

```

```

5   for (int m=1 ; m<n ; m<=1){
6       if (n<2*m) a.resize(2*m);
7       vector<int> v1(a.begin(), a.begin()+2*m), v2 = ret;
8       v1.resize(4*m), v2.resize(4*m);
9       NTT(v1, NTT(v2));
10      for (int i=0 ; i<4*m ; i++) v1[i] = v1[i]*v2[i]%MOD*
          v2[i]%MOD;
11      NTT(v1, true);
12      ret.resize(2*m);
13      for (int i=0 ; i<m ; i++) ret[i] = (ret[i]+ret[i])%
          MOD;
14      for (int i=0 ; i<2*m ; i++) ret[i] = ((ret[i]-v1[i])%
          MOD+MOD)%MOD;
15  }
16  ret.resize(n);
17  return ret;
18 }

```

2.7 PolySqrt [1ea321]

```

1  // ---- sqrt for formal power series over MOD=998244353 ----
2  int mod_sqrt(int a) { // Tonelli-Shanks; return -1 if no
    solution
3      a %= MOD; if (a < 0) a += MOD;
4      if (a == 0) return 0;
5      if (qp(a, (MOD - 1) / 2, MOD) == MOD - 1) return -1; //
        non-residue
6      if (MOD % 4 == 3) return qp(a, (MOD + 1) / 4, MOD);
7      int q = MOD - 1, s = 0;
8      while ((q & 1) == 0) q >>= 1, ++s;
9      int z = 2;
10     while (qp(z, (MOD - 1) / 2, MOD) != MOD - 1) ++z; // find
        non-residue
11     long long c = qp(z, q, MOD);
12     long long x = qp(a, (q + 1) / 2, MOD);
13     long long t = qp(a, q, MOD);
14     int m = s;
15     while (t != 1) {
16         int i = 1;
17         long long tt = t * t % MOD;
18         while (tt != 1) {
19             tt = tt * tt % MOD;
20             ++i;
21             if (i == m) return -1; // shouldn't happen
22         }
23         long long b = qp(c, 1LL << (m - i - 1), MOD);
24         x = x * b % MOD;
25         long long b2 = b * b % MOD;
26         t = t * b2 % MOD;
27         c = b2;
28         m = i;
29     }
30     return (int)x;
31 }
32
33 vector<int> polySqrt(vector<int> a) {
34     int n = (int)a.size();
35
36     // all zero -> zero
37     bool all0 = true;
38     for (int x : a) if (x % MOD != 0) { all0 = false; break;
39     }
40     if (all0) return vector<int>(n, 0);
41     // first nonzero position

```

```

42     int k = 0;
43     while (k < n && a[k] == 0) ++k;
44     if (k & 1) return {}; // odd -> impossible
45     int shift = k / 2;
46
47     // strip x^k
48     vector<int> f;
49     f.reserve(n - k);
50     for (int i = k; i < n; ++i) f.push_back(a[i]);
51
52     // constant term sqrt, choose canonical sign
53     int c0 = (f.empty() ? 0 : f[0] % MOD);
54     int s0 = mod_sqrt(c0);
55     if (s0 == -1) return {};
56     if (s0 > MOD - s0) s0 = MOD - s0; // choose the smaller
        nonnegative root
57
58     // We must produce N terms after placing the shift =>
        need = N - shift
59     int need = n - shift;
60     if ((int)f.size() < need) f.resize(need, 0);
61
62     // Newton: g_{new} = (g + f/g) / 2 (mod x^m)
63     const int INV2 = (MOD + 1) / 2;
64     vector<int> g(1, s0);
65
66     for (int m = 1; m < need; m <= 1) {
67         int lim = min(need, m << 1);
68
69         vector<int> g_cut = g; g_cut.resize(lim);
70         vector<int> inv_g = polyInv(g_cut); // inv up
            to lim
71         vector<int> f_cut(f.begin(), f.begin() + lim);
72
73         vector<int> q = polyMul(inv_g, f_cut);
74         if ((int)q.size() > lim) q.resize(lim);
75
76         g.resize(lim);
77         for (int i = 0; i < lim; ++i) {
78             long long gi = (i < (int)g_cut.size() ? g_cut[i]
                : 0);
79             long long qi = (i < (int)q.size() ? q[i] : 0);
80             g[i] = (gi + qi) % MOD * 1LL * INV2 % MOD;
81         }
82     }
83     g.resize(need);
84
85     // place back the x^{k/2} shift
86     vector<int> res(n, 0);
87     for (int i = 0; i < need && i + shift < n; ++i) res[i +
        shift] = g[i];
88     return res;
89 }

```

3 Data-Structure

3.1 BIT [7ef3a9]

```

1  vector<int> BIT(MAX_SIZE);
2
3  // const int MAX_N = (1<<20)
4  int k_th(int k) { // 回傳 BIT 中第 k 小的元素 (based-1)
5      int res = 0;
6      for (int i=MAX_N>>1 ; i>=1 ; i>=1)
7          if (BIT[res+i]<k)

```

```

8          k -= BIT[res+i];
9      return res+1;
10 }

```

3.2 Disjoint Set Persistent [447002]

```

1  struct Persistent_Disjoint_Set{
2      Persistent_Segment_Tree arr, sz;
3
4      void init(int n){
5          arr.init(n);
6          vector<int> v1;
7          for (int i=0 ; i<n ; i++){
8              v1.push_back(i);
9          }
10         arr.build(v1, 0);
11
12         sz.init(n);
13         vector<int> v2;
14         for (int i=0 ; i<n ; i++){
15             v2.push_back(1);
16         }
17         sz.build(v2, 0);
18     }
19
20     int find(int a){
21         int res = arr.query_version(a, a+1, arr.version.size()
            -1).val;
22         if (res==a) return a;
23         return find(res);
24     }
25
26     bool unite(int a, int b){
27         a = find(a);
28         b = find(b);
29
30         if (a!=b){
31             int sz1 = sz.query_version(a, a+1, arr.version.
                size()-1).val;
32             int sz2 = sz.query_version(b, b+1, arr.version.
                size()-1).val;
33
34             if (sz1<sz2){
35                 arr.update_version(a, b, arr.version.size()
                    -1);
36                 sz.update_version(b, sz1+sz2, arr.version.
                    size()-1);
37             }else{
38                 arr.update_version(b, a, arr.version.size()
                    -1);
39                 sz.update_version(a, sz1+sz2, arr.version.
                    size()-1);
40             }
41             return true;
42         }
43         return false;
44     }
45 }
46

```

3.3 PBDS GP Hash Table [866cf6]

```

1  #include <ext/pb_ds/assoc_container.hpp>
2  using namespace __gnu_pbds;

```

```

3 typedef tree<int,null_type,less<int>,rb_tree_tag,
  tree_order_statistics_node_update> order_set;
4 struct custom_hash {
5     static uint64_t splitmix64(uint64_t x) {
6         // http://xorshift.di.unimi.it/splitmix64.c
7         x += 0x9e3779b97f4a7c15;
8         x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
9         x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
10        return x ^ (x >> 31);
11    }
12
13    size_t operator()(uint64_t x) const {
14        static const uint64_t FIXED_RANDOM = chrono::
15        steady_clock::now().time_since_epoch().count();
16        return splitmix64(x + FIXED_RANDOM);
17    }
18 };
19 gp_hash_table<int, int, custom_hash> ss;

```

3.4 PBDS Order Set [231774]

```

1 // .find_by_order(k) 回傳第 k 小的值 (based-0)
2 // .order_of_key(k) 回傳有多少元素比 k 小
3 // 不能在 #define int long long 後 #include 檔案
4 #include <ext/pb_ds/assoc_container.hpp>
5 #include <ext/pb_ds/tree_policy.hpp>
6 using namespace __gnu_pbds;
7 typedef tree<int,null_type,less<int>,rb_tree_tag,
  tree_order_statistics_node_update> order_set;

```

3.5 Segment Tree Add Set [bb1898]

```

1 // [ll, rr), based-0
2 // 使用前記得 init(陣列大小), build(陣列名稱)
3 // add(ll, rr): 區間修改
4 // set(ll, rr): 區間賦值
5 // query(ll, rr): 區間求和 / 求最大值
6 struct SegmentTree{
7     struct node{
8         int add_tag = 0;
9         int set_tag = 0;
10        int sum = 0;
11        int ma = 0;
12    };
13
14    vector<node> arr;
15
16    SegmentTree(int n){
17        arr.resize(n<<2);
18    }
19
20    node pull(node A, node B){
21        node C;
22        C.sum = A.sum+B.sum;
23        C.ma = max(A.ma, B.ma);
24        return C;
25    }
26
27    // cce0c8
28    void push(int idx, int ll, int rr){
29        if (arr[idx].set_tag!=0){
30            arr[idx].sum = (rr-ll)*arr[idx].set_tag;
31            arr[idx].ma = arr[idx].set_tag;

```

```

32        if (rr-ll>1){
33            arr[idx*2+1].add_tag = 0;
34            arr[idx*2+1].set_tag = arr[idx].set_tag;
35            arr[idx*2+2].add_tag = 0;
36            arr[idx*2+2].set_tag = arr[idx].set_tag;
37        }
38        arr[idx].set_tag = 0;
39    }
40
41    if (arr[idx].add_tag!=0){
42        arr[idx].sum += (rr-ll)*arr[idx].add_tag;
43        arr[idx].ma += arr[idx].add_tag;
44        if (rr-ll>1){
45            arr[idx*2+1].add_tag += arr[idx].add_tag;
46            arr[idx*2+2].add_tag += arr[idx].add_tag;
47        }
48        arr[idx].add_tag = 0;
49    }
50
51    }
52
53    void build(vector<int> &v, int idx = 0, int ll = 0, int
  rr = n){
54        if (rr-ll==1){
55            arr[idx].sum = v[ll];
56            arr[idx].ma = v[ll];
57        }else{
58            int mid = (ll+rr)/2;
59            build(v, idx*2+1, ll, mid);
60            build(v, idx*2+2, mid, rr);
61            arr[idx] = pull(arr[idx*2+1], arr[idx*2+2]);
62        }
63    }
64
65    void add(int ql, int qr, int val, int idx = 0, int ll =
  0, int rr = n){
66        push(idx, ll, rr);
67        if (rr<=ql || qr<=ll) return;
68        if (ql<=ll && qr<=rr){
69            arr[idx].add_tag += val;
70            push(idx, ll, rr);
71            return;
72        }
73        int mid = (ll+rr)/2;
74        add(ql, qr, val, idx*2+1, ll, mid);
75        add(ql, qr, val, idx*2+2, mid, rr);
76        arr[idx]=pull(arr[idx*2+1], arr[idx*2+2]);
77    }
78
79    void set(int ql, int qr, int val, int idx=0, int ll=0,
  int rr=n){
80        push(idx, ll, rr);
81        if (rr<=ql || qr<=ll) return;
82        if (ql<=ll && rr<=qr){
83            arr[idx].add_tag = 0;
84            arr[idx].set_tag = val;
85            push(idx, ll, rr);
86            return;
87        }
88        int mid = (ll+rr)/2;
89        set(ql, qr, val, idx*2+1, ll, mid);
90        set(ql, qr, val, idx*2+2, mid, rr);
91        arr[idx] = pull(arr[idx*2+1], arr[idx*2+2]);
92    }
93
94    node query(int ql, int qr, int idx = 0, int ll = 0, int
  rr = n){
95        push(idx, ll, rr);

```

```

94        if (rr<=ql || qr<=ll) return node();
95        if (ql<=ll && rr<=qr) return arr[idx];
96
97        int mid = (ll+rr)/2;
98        return pull(query(ql, qr, idx*2+1, ll, mid), query(ql
  , qr, idx*2+2, mid, rr));
99    }
100 } ST;

```

3.6 Segment Tree Li Chao Line [45b8ba]

```

1 // 全部都是 0-based
2 // 宣告 : LC_Segment_Tree st(n);
3 // 函式 :
4 // update({a, b}) : 插入一條 y=ax+b 的全域直線
5 // query(x) : 查詢所有直線在位置 x 的最小值
6 const int MAX_V = 1e6+10; // 值域最大值
7
8 struct LC_Segment_Tree{
9     struct Node{ // y = ax+b
10        int a = 0;
11        int b = INF;
12
13        int y(int x){
14            return a*x+b;
15        }
16    };
17    vector<Node> arr;
18
19    LC_Segment_Tree(int n = 0){
20        arr.resize(4*n);
21    }
22
23    void update(Node val, int idx = 0, int ll = 0, int rr =
  MAX_V){
24        if (rr-ll==0) return;
25        if (rr-ll==1){
26            if (val.y(ll)<arr[idx].y(ll)){
27                arr[idx] = val;
28            }
29            return;
30        }
31
32        int mid = (ll+rr)/2;
33        if (arr[idx].a > val.a) swap(arr[idx], val); // 原本
  的線斜率要比較小
34        if (arr[idx].y(mid) < val.y(mid)){ // 交點在左邊
35            update(val, idx*2+1, ll, mid);
36        }else{ // 交點在右邊
37            swap(arr[idx], val); // 在左子樹中，新線比舊線還
  要好
38            update(val, idx*2+2, mid, rr);
39        }
40        return;
41    }
42
43    int query(int x, int idx = 0, int ll = 0, int rr = MAX_V)
  {
44        if (rr-ll==0) return INF;
45        if (rr-ll==1){
46            return arr[idx].y(ll);
47        }
48
49        int mid = (ll+rr)/2;

```

```

50     if (x<mid){
51         return min(arr[idx].y(x), query(x, idx*2+1, 1l,
52             mid));
53     }else{
54         return min(arr[idx].y(x), query(x, idx*2+2, mid,
55             rr));
56     }
57 }
58 };

```

3.7 Segment Tree Li Chao Segment [cb9b37]

```

1 // 全部都是  $\theta$ -based
2 // 宣告: LC_Segment_Tree st(n); // n = 值域大小 · x 的範圍為
3 // 函式:
4 // st.update_segment({a, b}, ql, qr): 在 [ql, qr) 插入一條  $y =$ 
5 //  $ax+b$  的線段
6 // st.query(x): 查詢所有線段在位置 x 的最小值
7 struct LC_Segment_Tree {
8     struct Line { //  $y = ax + b$ 
9         int a = 0, b = INF;
10        int y(int x) const { return a * x + b; }
11    };
12
13    int N;
14    vector<Line> arr;
15    LC_Segment_Tree(int n) : N(n), arr(4 * n + 5) {}
16
17    void insert_line(Line val, int idx, int l, int r) {
18        if (arr[idx].a > val.a) swap(arr[idx], val);
19        if (arr[idx].a == val.a) {
20            arr[idx].b = min(arr[idx].b, val.b);
21            return;
22        }
23        if (val.y(l) >= arr[idx].y(l)) return;
24        if (val.y(r - 1) < arr[idx].y(r - 1)) {
25            arr[idx] = val;
26            return;
27        }
28        int mid = (l + r) / 2;
29        if (val.y(mid) < arr[idx].y(mid)) {
30            swap(arr[idx], val);
31            insert_line(val, 2 * idx + 2, mid, r);
32        }
33        else insert_line(val, 2 * idx + 1, l, mid);
34    }
35
36    void dfs(Line v, int ql, int qr, int idx, int l, int r) {
37        if (l >= r || r <= ql || l >= qr) return;
38        if (ql <= l && r <= qr) {
39            insert_line(v, idx, l, r);
40            return;
41        }
42        int mid = (l + r) / 2;
43        dfs(v, ql, qr, 2 * idx + 1, l, mid);
44        dfs(v, ql, qr, 2 * idx + 2, mid, r);
45    }
46
47    int q(int x, int idx, int l, int r) {
48        if (l >= r) return INF;
49        int res = arr[idx].y(x);
50        if (l + 1 == r) return res;
51        int mid = (l + r) / 2;

```

```

52     if (x < mid) return min(res, q(x, 2*idx+1, l, mid));
53     else return min(res, q(x, 2*idx+2, mid, r));
54 }
55
56 void update_segment(Line val, int ql, int qr) {
57     dfs(val, ql, qr, 0, 0, N);
58 }
59 int query(int x) { return q(x, 0, 0, N); }
60 };

```

3.8 Segment Tree Persistent [67eafb]

```

1 // Each operation adds at most  $(K \times \text{ceil}(\log_2(n)))$  nodes
2 // Range update :  $K = 8$ , Range query :  $K = 4$ 
3 // Only point update :  $\text{ceil}(\log_2(n)) + 1$ 
4 // To optimize K, consider 懶標永久化
5 struct Persistent_Segment_Tree { // 1-based, [l, r]
6     struct Node {
7         int lc = 0, rc = 0, sum = 0, tag = 0;
8     };
9     vector<Node> tr;
10
11     Persistent_Segment_Tree() {
12         tr.push_back({}); // Node 0 is null dummy
13     }
14
15     void apply(int u, int len, int v) {
16         tr[u].sum += v * len;
17         tr[u].tag += v;
18     }
19
20     void pushup(Node& u) {
21         u.sum = tr[u.lc].sum + tr[u.rc].sum;
22     }
23
24     Node merge(const Node& a, const Node& b) const {
25         return {0, 0, a.sum + b.sum, 0};
26     }
27
28     void pushdown(int u, int l, int r) {
29         if (!tr[u].tag) return;
30         int mid = (l + r) >> 1, tag = tr[u].tag;
31         int lc = tr.size(), rc = tr.size() + 1;
32         tr.push_back(tr[tr[u].lc]); tr.push_back(tr[tr[u].rc]);
33         apply(lc, mid - l + 1, tag); apply(rc, r - mid, tag);
34         tr[u].tag = 0; tr[u].lc = lc; tr[u].rc = rc;
35     }
36
37     // Must build before use, returns root index
38     int build(int l, int r, const vector<int>& a = {}) {
39         int u = tr.size(); tr.push_back({});
40         if (l == r) {
41             tr[u].sum = (a.empty() ? 0 : a[l]);
42             return u;
43         }
44         int mid = (l + r) >> 1;
45         int lc = build(l, mid, a), rc = build(mid + 1, r, a);
46         tr[u].lc = lc; tr[u].rc = rc;
47         pushup(tr[u]);
48         return u;
49     }
50
51     // Returns root index
52     int update(int pre, int l, int r, int ql, int qr, int v) {
53         int u = tr.size(); tr.push_back(tr[pre]);

```

```

54     if (ql <= l && r <= qr) {
55         apply(u, r - l + 1, v);
56         return u;
57     }
58     pushdown(u, l, r);
59     int mid = (l + r) >> 1, lc = tr[u].lc, rc = tr[u].rc;
60     if (ql <= mid) lc = update(lc, l, mid, ql, qr, v);
61     if (qr > mid) rc = update(rc, mid + 1, r, ql, qr, v);
62     tr[u].lc = lc; tr[u].rc = rc;
63     pushup(tr[u]);
64     return u;
65 }
66
67 Node query(int u, int l, int r, int ql, int qr) {
68     if (ql <= l && r <= qr) return tr[u];
69     pushdown(u, l, r);
70     int mid = (l + r) >> 1, lc = tr[u].lc, rc = tr[u].rc;
71     if (qr <= mid) return query(lc, l, mid, ql, qr);
72     if (ql > mid) return query(rc, mid + 1, r, ql, qr);
73     return merge(query(lc, l, mid, ql, qr),
74         query(rc, mid + 1, r, ql, qr));
75 }
76 };

```

3.9 Segment Tree Zkw [69ce93]

```

1 struct SegmentTreeZkw { // 1-based index
2     using T = array<int, 2>;
3     using U = int;
4     int n, m;
5     vector<T> seg;
6     vector<U> lazy;
7     static constexpr T empty_node = {0, 0};
8     static constexpr U empty_lazy = 1;
9
10    SegmentTreeZkw(int sz) : n(sz), m(sz - 1) {
11        seg.assign(2 * n, empty_node);
12        lazy.assign(2 * n, empty_lazy);
13    }
14    void build(T* a) {
15        for (int i = 1; i <= n; ++i) {
16            seg[m + i] = a[i];
17        }
18        for (int i = m; i >= 1; --i) {
19            seg[i] = merge(seg[i << 1], seg[i << 1 | 1]);
20        }
21    }
22    void range_update(int l, int r, U val) {
23        l += m; r += m;
24        int l0 = l, r0 = r;
25        push(l0), push(r0);
26        while (l <= r) {
27            if (l & 1) modify(l++, val);
28            if (!(r & 1)) modify(r--, val);
29            l >>= 1; r >>= 1;
30        }
31        push(l0), push(r0);
32        pull(l0), pull(r0);
33    }
34    void modify(int i, U val) {
35        seg[i][1] = seg[i][1] * val % mod;
36        lazy[i] = lazy[i] * val % mod;
37    }
38    void push(int i) {
39        for (int h = __lg(i); h >= 1; h--) {

```

```

40     int idx = i >> h;
41     if (lazy[idx] == empty_lazy) continue;
42     modify(idx << 1, lazy[idx]);
43     modify(idx << 1 | 1, lazy[idx]);
44     lazy[idx] = empty_lazy;
45 }
46 }
47 void pull(int i) {
48     while (i >= 1) {
49         seg[i] = merge(seg[i << 1], seg[i << 1 | 1]);
50     }
51 }
52 T merge(T x, T y) {
53     return {x[0] + y[0], (x[1] + hmul_pow[x[0]] * y[1]) %
54         mod};
55 }
56 T query(int l, int r) { // [l, r]
57     T resl = empty_node, resr = empty_node;
58     l += m; r += m;
59     push(l), push(r);
60     while (l <= r) {
61         if (l & 1) resl = merge(resl, seg[l++]);
62         if (!(r & 1)) resr = merge(seg[r--], resr);
63         l >>= 1; r >>= 1;
64     }
65     return merge(resl, resr);
66 };

```

3.10 Sparse Table [31f22a]

```

1 struct SparseTable{
2     vector<vector<int>> st;
3     void build(vector<int> v){
4         int h = __lg(v.size());
5         st.resize(h+1);
6         st[0] = v;
7
8         for (int i=1; i<=h; i++){
9             int gap = (1<<(i-1));
10            for (int j=0; j+gap<st[i-1].size(); j++){
11                st[i].push_back(min(st[i-1][j], st[i-1][j+gap]));
12            }
13        }
14    }
15
16    // 回傳 [ll, rr) 的最小值
17    int query(int ll, int rr){
18        int h = __lg(rr-ll);
19        return min(st[h][ll], st[h][rr-(1<<h)]);
20    }
21 };

```

3.11 Treap2 [3b0cca]

```

1 // 1-based · 請注意 MAX_N 是否足夠大
2 int root = 0;
3 int lc[MAX_N], rc[MAX_N];
4 int pri[MAX_N], val[MAX_N];
5 int sz[MAX_N], tag[MAX_N], fa[MAX_N], total[MAX_N];
6 // tag 為不包含自己 (僅要給子樹) 的資訊
7 int nodeCnt = 0;
8 int& new_node(int v){

```

```

9     nodeCnt++;
10    val[nodeCnt] = v;
11    total[nodeCnt] = v;
12    sz[nodeCnt] = 1;
13    pri[nodeCnt] = rand();
14    return nodeCnt;
15 }
16
17 void apply(int x, int V){
18     val[x] += V;
19     tag[x] += V;
20     total[x] += V*sz[x];
21 }
22
23 void push(int x){
24     if (tag[x]){
25         if (lc[x]) apply(lc[x], tag[x]);
26         if (rc[x]) apply(rc[x], tag[x]);
27     }
28     tag[x] = 0;
29 }
30
31 int pull(int x){
32     if (x){
33         fa[x] = 0;
34         sz[x] = 1+sz[lc[x]]+sz[rc[x]];
35         total[x] = val[x]+total[lc[x]]+total[rc[x]];
36         if (lc[x]) fa[lc[x]] = x;
37         if (rc[x]) fa[rc[x]] = x;
38     }
39     return x;
40 }
41
42 int merge(int a, int b){
43     if (!a or !b) return a|b;
44     push(a), push(b);
45
46     if (pri[a]>pri[b]){
47         rc[a] = merge(rc[a], b);
48         return pull(a);
49     }else{
50         lc[b] = merge(a, lc[b]);
51         return pull(b);
52     }
53 }
54
55 // [1, k] [k+1, n]
56 void split(int x, int k, int &a, int &b) {
57     if (!x) return a = b = 0, void();
58     push(x);
59     if (sz[lc[x]] >= k) {
60         split(lc[x], k, a, lc[x]);
61         b = x;
62         pull(a); pull(b);
63     }else{
64         split(rc[x], k - sz[lc[x]] - 1, rc[x], b);
65         a = x;
66         pull(a); pull(b);
67     }
68 }
69
70 // functions
71 // 回傳 x 在 Treap 中的位置
72 int get_pos(int x){
73     vector<int> sta;
74     while (fa[x]){

```

```

74         sta.push_back(x);
75         x = fa[x];
76     }
77     while (sta.size()){
78         push(x);
79         x = sta.back();
80         sta.pop_back();
81     }
82     push(x);
83
84     int res = sz[x] - sz[rc[x]];
85     while (fa[x]){
86         if (rc[fa[x]]==x){
87             res += sz[fa[x]]-sz[x];
88         }
89         x = fa[x];
90     }
91     return res;
92 }
93
94 // 1-based <前 [1, l-1] 個元素, [l, r] 個元素, [r+1, n] 個元
95 // 素>
96 array<int, 3> cut(int x, int l, int r){
97     array<int, 3> ret;
98     split(x, r, ret[1], ret[2]);
99     split(ret[1], l-1, ret[0], ret[1]);
100    return ret;
101 }
102
103 void print(int x){
104     push(x);
105     if (lc[x]) print(lc[x]);
106     cerr << val[x] << " ";
107     if (rc[x]) print(rc[x]);
108 }

```

3.12 Trie [b6475c]

```

1 struct Trie{
2     struct Data{
3         int nxt[2]={0, 0};
4     };
5
6     int sz=0;
7     vector<Data> arr;
8
9     void init(int n){
10        arr.resize(n);
11    }
12
13    void insert(int n){
14        int now=0;
15        for (int i=N; i>=0; i--){
16            int v=(n>>i)&1;
17            if (!arr[now].nxt[v]){
18                arr[now].nxt[v]=++sz;
19            }
20            now=arr[now].nxt[v];
21        }
22    }
23
24    int query(int n){
25        int now=0, ret=0;
26        for (int i=N; i>=0; i--){
27            int v=(n>>i)&1;

```

```

28         if (arr[now].nxt[1-v]){
29             ret+=(1<<i);
30             now=arr[now].nxt[1-v];
31         }else if (arr[now].nxt[v]){
32             now=arr[now].nxt[v];
33         }else{
34             return ret;
35         }
36     }
37     return ret;
38 }
39 } tr;
40 }

```

4 Dynamic-Programming

4.1 Digit DP [133f00]

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 long long l, r;
5 long long dp[20][10][2][2]; // dp[pos][pre][Limit] = 後 pos
   位 · pos 前一位是 pre · (是/否) 有上界 · (是/否) 有前綴零
   的答案數量
6
7 long long memorize_search(string &s, int pos, int pre, bool
   limit, bool lead){
8
9     // 已經被找過了 · 直接回傳值
10    if (dp[pos][pre][limit][lead]!=-1) return dp[pos][pre][
   limit][lead];
11
12    // 已經搜尋完畢 · 紀錄答案並回傳
13    if (pos==(int)s.size()){
14        return dp[pos][pre][limit][lead] = 1;
15    }
16
17    // 枚舉目前的位數數字是多少
18    long long ans = 0;
19    for (int now=0 ; now<=(limit ? s[pos]-'0' : 9) ; now++){
20        if (now==pre){
21
22            // 1~9 絕對不能連續出現
23            if (pre!=0) continue;
24
25            // 如果已經不在前綴零的範圍內 · 0 不能連續出現
26            if (lead==false) continue;
27        }
28
29        ans += memorize_search(s, pos+1, now, limit&(now==(s[
   pos]-'0')), lead&(now==0));
30    }
31
32    // 已經搜尋完畢 · 紀錄答案並回傳
33    return dp[pos][pre][limit][lead] = ans;
34 }
35
36 // 回傳 [0, n] 有多少數字符合條件
37 long long find_answer(long long n){
38     memset(dp, -1, sizeof(dp));
39     string tmp = to_string(n);
40
41     return memorize_search(tmp, 0, 0, true, true);

```

```

42 }
43
44 int main(){
45
46     // input
47     cin >> l >> r;
48
49     // output - 計算 [l, r] 有多少數字任意兩個位數都不相同
50     cout << find_answer(r)-find_answer(l-1) << "\n";
51
52     return 0;
53 }

```

4.2 Integer Partition

$dp[i][x]$ = 要將整數 x 拆成 i 堆的「組合數」

$dp[i+1][x+1] += dp[i][x]$ (創造新的一堆)
 $dp[i][x+i] += dp[i][x]$ (把每一堆都增加1)

4.3 Knapsack On Tree [df69b1]

```

1 // 需要重構、需要增加註解
2 #include <bits/stdc++.h>
3 #define F first
4 #define S second
5 #define all(x) begin(x), end(x)
6 using namespace std;
7
8 #define chmax(a, b) (a) = (a) < (b) ? (b) : (a)
9 #define chmin(a, b) (a) = (a) < (b) ? (a) : (b)
10
11 #define ll long long
12
13 #define FOR(i, a, b) for (int i = a; i <= b; i++)
14
15 int N, W, cur;
16 vector<int> w, v, sz;
17 vector<vector<int>> adj, dp;
18
19 void dfs(int x) {
20     sz[x] = 1;
21     for (int i : adj[x]) dfs(i), sz[x] += sz[i];
22     cur++;
23     // choose x
24     for (int i=w[x] ; i<=W ; i++){
25         dp[cur][i] = dp[cur - 1][i - w[x]] + v[x];
26     }
27     // not choose x
28     for (int i=0 ; i<=W ; i++){
29         chmax(dp[cur][i], dp[cur - sz[x]][i]);
30     }
31 }
32
33 signed main() {
34     cin >> N >> W;
35     adj.resize(N + 1);
36     w.assign(N + 1, 0);
37     v.assign(N + 1, 0);
38     sz.assign(N + 1, 0);
39     dp.assign(N + 2, vector<int>(W + 1, 0));
40     for (int i=1 ; i<=N ; i++){
41         int p; cin >> p;
42         adj[p].push_back(i);
43     }
44 }

```

```

45     for (int i=1 ; i<=N ; i++) cin >> w[i];
46     for (int i=1 ; i<=N ; i++) cin >> v[i];
47     dfs(0);
48     cout << dp[N + 1][W] << '\n';
49 }

```

4.4 SOS DP [8dfa8b]

```

1 // 總時間複雜度為  $O(n \cdot 2^n)$ 
2 // 計算  $dp[i] = i$  所有 bit mask 子集的和
3 for (int i=0 ; i<n ; i++){
4     for (int mask=0 ; mask<(1<<n) ; mask++){
5         if ((mask>>i)&1){
6             dp[mask] += dp[mask^(1<<i)];
7         }
8     }
9 }

```

5 Geometry

5.1 misc [3ef351]

```

1 using ld = double;
2
3 // 判斷數值正負：{1:正數,0:零,-1:負數}
4 int sign(long long x) {return (x >= 0) ? ((bool)x) : -1; }
5 int sign(ld x) {return (abs(x) < 1e-9) ? 0 : (x>0 ? 1 : -1);}

```

5.2 convex hull [bbcaaf]

```

1 int sign(long long x) {return (x >= 0) ? ((bool)x) : -1; }
2
3 template<typename T>
4 struct point {
5     T x, y;
6     point() {}
7     point(const T &x0, const T &y0) : x(x0), y(y0) {}
8
9     point operator-(point b) {return {x-b.x, y-b.y}; }
10    bool operator==(point b) {return x==b.x && y==b.y; }
11    T operator^(point b) {return x * b.y - y * b.x; }
12    friend int ori(point a, point b, point c) {
13        return sign((b-a)^(c-a));
14    }
15 };
16
17 template<typename T>
18 struct polygon {
19     vector<point<T>> v;
20     void make_convex_hull(int simple) {
21         auto cmp = [&](point<T> &p, point<T> &q) {
22             return (p.x == q.x) ? (p.y < q.y) : (p.x < q.x);
23         };
24         simple = (bool)simple;
25         sort(v.begin(), v.end(), cmp);
26         v.resize(unique(v.begin(), v.end()) - v.begin());
27         if (v.size() <= 1) return;
28         vector<point<T>> hull;
29         for (int t = 0; t < 2; ++t){
30             int sz = hull.size();
31             for (auto &i:v) {
32                 while (hull.size() >= sz+2 && ori(hull[hull.
33                     size()-2], hull.back(), i) < simple) {
34                     hull.pop_back();

```

```

35         }
36         hull.push_back(i);
37     }
38     hull.pop_back();
39     reverse(v.begin(), v.end());
40 }
41 swap(hull, v);
42 }
43 };

```

5.3 SVG writer [5bf313]

```

1 class SVG {
2     void p(string_view s) { o << s; }
3     void p(string_view s, auto v, auto... vs) {
4         auto i = s.find('$');
5         o << s.substr(0, i) << v, p(s.substr(i + 1), vs...);
6     }
7     ofstream o;
8     string c = "red";
9 public:
10    SVG(auto f, auto x1, auto y1, auto x2, auto y2) : o(f) {
11        p("<svg xmlns='http://www.w3.org/2000/svg' "
12        "viewBox='$ $ $ $'>\n"
13        "<style>{*{stroke-width:0.5%;}</style>\n",
14        x1, -y2, x2 - x1, y2 - y1);
15    } // cc67e3
16    ~SVG() { p("</svg>\n"); }
17    void color(string nc) { c = nc; }
18    void line(auto x1, auto y1, auto x2, auto y2) {
19        p("<line x1='$' y1='$' x2='$' y2='$' stroke='$'>\n",
20        x1, -y1, x2, -y2, c);
21    } // a08d95
22    void circle(auto x, auto y, auto r) {
23        p("<circle cx='$' cy='$' r='$' stroke='$' "
24        "fill='$'>\n", x, -y, r, c, "#none");
25    } // 73c400
26    void text(auto x, auto y, string s, int w = 12) {
27        p("<text x='$' y='$' font-size='$px'>${</text>\n",
28        x, -y, w, s);
29    } // a23944
30 };
31 // declare : SVG svg("test.svg", -99, -99, 100, 100);

```

5.4 struct point [f21f07]

```

1 template<typename T>
2 struct point {
3     T x, y;
4     point() {}
5     point(const T &x0, const T &y0) : x(x0), y(y0) {}
6     explicit operator point<ld>() {return point<ld>(x, y); }
7 };

```

5.5 point / basic operation [c415da]

```

1 point operator+(point b) {return {x+b.x, y+b.y}; }
2 point operator-(point b) {return {x-b.x, y-b.y}; }
3 point operator*(T b) {return {x*b, y*b}; }
4 point operator/(T b) {return {x/b, y/b}; }
5 bool operator==(point b) {return x==b.x && y==b.y; }
6
7 T operator*(point b) {return x * b.x + y * b.y; }
8 T operator^(point b) {return x * b.y - y * b.x; }

```

5.6 point / operators [fc43e4]

```

1 // 逆時針極角排序
2 bool side() const{return (y == 0) ? (x > 0) : (y < 0); }
3 bool operator<(const point &b) const {
4     return side() == b.side() ?
5         (x*b.y > b.x*y) : side() < b.side();
6 }
7 friend ostream& operator<<(ostream& os, point p) {
8     return os << "(" << p.x << ", " << p.y << ")";
9 }
10 // 判斷 ab 到 ac 的方向 : {1:逆時鐘,0:重疊,-1:順時鐘}
11 friend int ori(point a, point b, point c) {
12     return sign((b-a)^(c-a));
13 }
14 friend bool btw(point a, point b, point c) {
15     return ori(a, b, c) == 0 && sign((a-c)*(b-c)) <= 0;
16 }

```

5.7 point / banana [09fd7c]

```

1 // 判斷線段 ab, cd 是否相交
2 friend bool banana(point a, point b, point c, point d) {
3     if (btw(a, b, c) || btw(a, b, d)
4         || btw(c, d, a) || btw(c, d, b)) return true;
5     int u = ori(a, b, c) * ori(a, b, d);
6     int v = ori(c, d, a) * ori(c, d, b);
7     return u < 0 && v < 0;
8 }

```

5.8 point / rayHitSeg [db541a]

```

1 // 判斷 "射線 ab" 與 "線段 cd" 是否相交
2 friend bool rayHitSeg(point a, point b, point c, point d) {
3     if (a == b) return btw(c, d, a); // Special case
4     if (((a - b) ^ (c - d)) == 0) {
5         return btw(a, c, b) || btw(a, d, b) || banana(a, b, c, d);
6     }
7     point u = b - a, v = d - c, s = c - a;
8     return sign(s ^ v) * sign(u ^ v) >= 0 && sign(s ^ u)
9         * sign(u ^ v) >= 0 && abs(s ^ u) <= abs(u ^ v);
10 }

```

5.9 point / else [f46547]

```

1 // 旋轉 Arg(b) 的角度 (小心溢位)
2 point rotate(point b){return {x*b.x-y*b.y, x*b.y+y*b.x};}
3 // 回傳極座標角度・值域: [-π, +π]
4 friend ld Arg(point b) {
5     return (b.x != 0 || b.y != 0) ? atan2(b.y, b.x) : 0;
6 }
7 friend T abs2(point b) {return b * b; }

```

5.10 struct line [210dc8]

```

1 template<typename T>
2 struct line {
3     point<T> p1, p2;
4     // ax + by + c = 0
5     T a, b, c; // |a|, |b| ≤ 2C, |c| ≤ 8C²
6     line() {}
7     line(const point<T> &x, const point<T> &y) : p1(x), p2(y){

```

```

8         build();
9     }
10    void build() {
11        a = p1.y - p2.y;
12        b = p2.x - p1.x;
13        c = (-a*p1.x)-b*p1.y;
14    }
15 };

```

5.11 line / else [09bcc9]

```

1 // 判斷點和有向直線的關係 : {1:左邊,0:在線上,-1:右邊}
2 int ori(point<T> &p) {
3     return sign((p2-p1) ^ (p-p1));
4 }
5 // 判斷直線斜率是否相同
6 bool parallel(line &l) {
7     return ((p1-p2) ^ (l.p1-l.p2)) == 0;
8 }
9 // 兩直線交點
10 point<ld> line_intersection(line &l) {
11     using P = point<ld>;
12     point<T> u = p2-p1, v = l.p2-l.p1, s = l.p1-p1;
13     return P(p1) + P(u) * ((ld(s^v)) / (u^v));
14 }

```

5.12 struct polygon [171a85]

```

1 template<typename T>
2 struct polygon {
3     vector<point<T>> v;
4     polygon() {}
5     polygon(const vector<point<T>> &u) : v(u) {}
6 };

```

5.13 polygon / make convex hull [2bb3ef]

```

1 // simple 為 true 的時候會回傳任意三點不共線的凸包
2 void make_convex_hull(int simple) {
3     auto cmp = [&](point<T> &p, point<T> &q) {
4         return (p.x == q.x) ? (p.y < q.y) : (p.x < q.x);
5     };
6     simple = (bool)simple;
7     sort(v.begin(), v.end(), cmp);
8     v.resize(unique(v.begin(), v.end()) - v.begin());
9     if (v.size() <= 1) return;
10    vector<point<T>> hull;
11    for (int t = 0; t < 2; ++t){
12        int sz = hull.size();
13        for (auto &i:v) {
14            while (hull.size() >= sz+2 && ori(hull[hull.
15                size()-2], hull.back(), i) < simple) {
16                hull.pop_back();
17            }
18            hull.push_back(i);
19        }
20        hull.pop_back();
21        reverse(v.begin(), v.end());
22    }
23    swap(hull, v);
24 }

```

5.14 polygon / in polygon [f11340]

```

1 // 可以在有  $n$  個點的簡單多邊形內，用  $O(n)$  判斷一個點：
2 // {1 : 在多邊形內, 0 : 在多邊形上, -1 : 在多邊形外}
3 int in_polygon(point<T> a) {
4     const T MAX_POS = 1e9 + 5; // [記得修改] 座標的最大值
5     point<T> pre = v.back(), b(MAX_POS, a.y + 1);
6     int cnt = 0;
7
8     for (auto &i:v) {
9         if (btw(pre, i, a)) return 0;
10        if (banana(a, b, pre, i)) cnt++;
11        pre = i;
12    }
13
14    return cnt%2 ? 1 : -1;
15 }

```

5.15 polygon / in convex [e64f1e]

```

1 /// 警告：以下所有凸包專用的函式都只接受逆時針排序且任三點不
2 共線的凸包 ///
3 // 可以在有  $n$  個點的凸包內，用  $O(\log n)$  判斷一個點：
4 // {1 : 在凸包內, 0 : 在凸包邊上, -1 : 在凸包外}
5 int in_convex(point<T> p) {
6     int n = v.size();
7     int a = ori(v[0], v[1], p), b = ori(v[0], v[n-1], p);
8     if (a < 0 || b > 0) return -1;
9     if (btw(v[0], v[1], p)) return 0;
10    if (btw(v[0], v[n-1], p)) return 0;
11    int l = 1, r = n - 1, mid;
12    while (l + 1 < r) {
13        mid = (l + r) >> 1;
14        if (ori(v[0], v[mid], p) >= 0) l = mid;
15        else r = mid;
16    }
17    int k = ori(v[l], v[r], p);
18    if (k <= 0) return k;
19    return 1;
20 }

```

5.16 polygon / cycle search [fe2f51]

```

1 // 凸包專用的環狀二分搜，回傳  $\theta$ -based index
2 int cycle_search(auto &f) {
3     int n = v.size(), l = 0, r = n;
4     if (n == 1) return 0;
5     bool rv = f(1, 0);
6     while (r - l > 1) {
7         int m = (l + r) / 2;
8         if (f(0, m) ? rv: f(m, (m + 1) % n)) r = m;
9         else l = m;
10    }
11    return f(1, r % n) ? 1 : r % n;
12 }

```

5.17 polygon / line cut convex [b6a4c8]

```

1 // 可以在有  $n$  個點的凸包內，用  $O(\log n)$  判斷一條直線：
2 // {1 : 穿過凸包, 0 : 剛好切過凸包, -1 : 沒碰到凸包}
3 int line_cut_convex(line<T> L) {
4     L.build();
5     point<T> p(L.a, L.b);

```

```

6     auto gt = [&](int neg) {
7         auto f = [&](int x, int y) {
8             return sign((v[x] - v[y]) * p) == neg;
9         };
10        return -(v[cycle_search(f)] * p);
11    };
12    T x = gt(1), y = gt(-1);
13    if (L.c < x || y < L.c) return -1;
14    return not (L.c == x || L.c == y);
15 }

```

5.18 polygon / convex line intersect [47e699]

```

1 // return edge endpoint index
2 // but if (first == second) => intersect at a vertex
3 vector<pair<int,int>> convex_line_intersect(line<T> L) {
4     auto dis = [&](int p) {
5         return (L.p2 - L.p1) ^ (v[p] - L.p1);
6     };
7     auto gao = [&](int s) {
8         auto f = [&](int i, int j) {
9             return sign(dis(i) - dis(j)) == s;
10        };
11        return cycle_search(f);
12    };
13    int x = gao(1), y = gao(-1), n = v.size();
14    if (sign(dis(x)) < 0 || sign(dis(y)) > 0) return {};
15    if (sign(dis(x)) == 0 || sign(dis(y)) == 0) {
16        int j = ((sign(dis(x)) == 0 ? x : y) + n - 1) % n;
17        vector<pair<int,int>> ret;
18        for (int i = 0; i < 3; ++i, j = (j + 1) % n)
19            if (sign(dis(j)) == 0) ret.emplace_back(j, j);
20        return ret;
21    }
22    auto g = [&](int l, int r, int s) -> pair<int,int> {
23        while ((l + 1) % n != r) {
24            int m = ((l + r + (l < r ? 0 : n)) / 2) % n;
25            if (sign(dis(m)) == s ? l : r) = m;
26        }
27        return {sign(dis(r)) ? l : r, r};
28    };
29    return {g(x, y, 1), g(y, x, -1)};
30 }

```

5.19 polygon / segment pass convex interior [7a3f22]

```

1 // 可以在有  $n$  個點的凸包內，用  $O(\log n)$  判斷一個線段：
2 // {1 : 線段上存在某一點位於凸包內部（邊上不算），
3 // 0 : 線段上存在某一點碰到凸包的邊但線段上任一點均不在凸包
4 // 內部，
5 // -1 : 線段完全在凸包外面}
6 int segment_pass_convex_interior(line<T> L) {
7     if (in_convex(L.p1) == 1 || in_convex(L.p2) == 1) return
8         1;
9     L.build();
10    auto res = convex_line_intersect(L);
11    if (res.empty()) return -1;
12    bool cntp = 0, cntn = 0;
13    for (auto &[i, j] : res) if (banana(v[i], v[j], L.p1, L.
14        p2)) {
15        int now = L.ori(v[(i + 1) % v.size()]);
16        cntp |= (now == 1);
17        cntn |= (now == -1);
18    }

```

```

16    return (-1 + cntp + cntn);
17 }

```

5.20 polygon / convex tangent point [a6f66b]

```

1 // 回傳點過凸包的兩條切線的切點的  $\theta$ -based index (不保證兩條
2 切線的順逆時針關係)
3 pair<int,int> convex_tangent_point(point<T> p) {
4     int n = v.size(), z = -1, edg = -1;
5     auto gt = [&](int neg) {
6         auto check = [&](int x) {
7             if (v[x] == p) z = x;
8             if (btw(v[x], v[(x + 1) % n], p)) edg = x;
9             if (btw(v[(x + n - 1) % n], v[x], p)) edg = (x +
10                 n - 1) % n;
11         };
12         auto f = [&](int x, int y) {
13             check(x); check(y);
14             return ori(p, v[x], v[y]) == neg;
15         };
16         return cycle_search(f);
17     };
18    int x = gt(1), y = gt(-1);
19    if (z != -1) {
20        return {(z + n - 1) % n, (z + 1) % n};
21    }
22    else if (edg != -1) {
23        return {edg, (edg + 1) % n};
24    }
25    else {
26        return {x, y};
27    }

```

5.21 polygon / halfplane intersection [102d48]

```

1 friend int halfplane_intersection(vector<line<T>> &s, polygon
2 <T> &P) {
3     auto angle_cmp = [&](line<T> &A, line<T> &B) {
4         point<T> a = A.p2-A.p1, b = B.p2-B.p1;
5         return (a < b);
6     };
7     sort(s.begin(), s.end(), angle_cmp); // 線段左側為該線段
8     半平面
9     int L, R, n = s.size();
10    vector<point<T>> px(n);
11    vector<line<T>> q(n);
12    q[L = R = 0] = s[0];
13    for (int i = 1; i < n; ++i) {
14        while (L < R && s[i].ori(px[R-1]) <= 0) --R;
15        while (L < R && s[i].ori(px[L]) <= 0) ++L;
16        q[++R] = s[i];
17        if (q[R].parallel(q[R-1])) {
18            --R;
19            if (q[R].ori(s[i].p1) > 0) q[R] = s[i];
20        }
21        if (L < R) px[R-1] = q[R-1].line_intersection(q[R]);
22    }
23    while (L < R && q[L].ori(px[R-1]) <= 0) --R;
24    P.v.clear();
25    if (R - L <= 1) return 0;
26    px[R] = q[R].line_intersection(q[L]);
27    for (int i = L; i <= R; ++i) P.v.push_back(px[i]);
28    return R - L + 1;
29 }

```

5.22 polygon / minkowski sum [aba519]

```

1 void minkowski(vector<point<T>> a, vector<point<T>> b) {
2     // a, b: convex polygon's inside vector, CCW order
3     auto cmp = [&](auto &a, auto &b) {
4         return tie(a.y, a.x) < tie(b.y, b.x);
5     };
6     auto reorder = [&](auto &u) {
7         auto it = min_element(u.begin(), u.end(), cmp);
8         rotate(u.begin(), it, u.end());
9     };
10    reorder(a); reorder(b);
11    a.push_back(a[0]); a.push_back(a[1]);
12    b.push_back(b[0]); b.push_back(b[1]);
13    v.clear();
14    for (int i = 0, j = 0; i+2<a.size()||j+2<b.size();) {
15        v.push_back(a[i] + b[j]);
16        auto val = (a[i + 1] - a[i]) ^ (b[j + 1] - b[j]);
17        if (val >= 0) i++;
18        if (val <= 0) j++;
19    }
20 }

```

5.23 struct Cir [87198d]

```

1 struct Cir {
2     point<ld> o; ld r;
3     friend ostream& operator<<(ostream& os, Cir c) {
4         return os << "(x" << "+-" [c.o.x >= 0] << abs(c.o.x)
5             << ")^2 + (y" << "+-" [c.o.y >= 0] << abs(c.o.y)
6             << ")^2 = " << c.r * c.r;
7     }
8     bool covers(Cir b) {
9         return sqrt((ld)abs2(o - b.o)) + b.r <= r;
10    }
11 };

```

5.24 Cir / Cir intersect [330a1c]

```

1 vector<point<ld>> Cir_intersect(Cir c) {
2     ld d2 = abs2(o - c.o), d = sqrt(d2);
3     if (d < max(r, c.r) - min(r, c.r) || d > r + c.r) return {};
4     auto sqdf = [&](ld x, ld y) { return x*x - y*y; };
5     point<ld> u = (o + c.o) / 2 + (o - c.o) * (sqdf(c.r, r) / (2 * d2));
6     ld A = sqrt(sqdf(r + d, c.r) * sqdf(c.r, d - r));
7     point<ld> v = (c.o - o).rotate({0,1}) * A / (2 * d2);
8     if (sign(v.x) == 0 && sign(v.y) == 0) return {u};
9     return {u - v, u + v};
10 }

```

5.25 Cir / point tangent [0067e6]

```

1 auto point_tangent(point<ld> p) {
2     vector<point<ld>> res;
3     ld d_sq = abs2(p - o);
4     if (sign(d_sq - r * r) <= 0) {
5         res.pb(p + (p - o).rotate({0, 1}));
6     } else if (d_sq > r * r) {
7         ld s = d_sq - r * r;
8         point<ld> v = p + (o - p) * s / d_sq;
9         point<ld> u = (o - p).rotate({0, 1}) * sqrt(s) * r / d_sq;
10    }

```

```

10    res.pb(v + u);
11    res.pb(v - u);
12 }
13 return res;
14 }

```

5.26 Cir / circle tangent [a3c7fb]

```

1 // returns nothing when two circles are the same
2 auto circle_tangent(Cir c2, int sign1) {
3     // sign1 = 1 for outer tang, -1 for inter tang
4     using ptld = point<ld>;
5     vector<pair<ptld, ptld>> res;
6     ld d_sq = abs2(o - c2.o);
7     if (sign(d_sq) == 0) return res;
8     ld d = sqrt(d_sq);
9     ptld v = (c2.o - o) / d;
10    ld c = (r - sign1 * c2.r) / d;
11    if (c * c > 1) return res;
12    ld h = sqrt(max((ld)0.0, 1.0 - c * c));
13    for (int sign2 = 1; sign2 >= -1; sign2 -= 2) {
14        ptld n(v.x * c - sign2 * h * v.y, v.y * c + sign2 * h
15            * v.x);
16        ptld p1 = o + n * r;
17        ptld p2 = c2.o + n * (c2.r * sign1);
18        if (sign(p1.x - p2.x) == 0 && sign(p1.y - p2.y) == 0)
19            p2 = p1 + (c2.o - o).rotate({0, 1});
20        res.push_back({p1, p2});
21    }
22    if (sign1 == -1 && sign(r + c2.r - d) == 0) res.pop_back();
23    if (sign1 == 1 && sign(abs(r - c2.r) - d) == 0) res.pop_back();
24    return res;
25 }

```

5.27 Cir / minimum enclosing circle [3e1440]

```

1 point<ld> circenter(point<ld> p0, point<ld> p1, point<ld> p2) {
2     // radius = abs(center)
3     p1 = p1 - p0, p2 = p2 - p0;
4     ld x1 = p1.x, y1 = p1.y, x2 = p2.x, y2 = p2.y;
5     ld m = 2. * (x1 * y2 - y1 * x2);
6     point<ld> center(0, 0);
7     center.x = (x1 * x1 * y2 - x2 * x2 * y1 + y1 * y2 * (y1 - y2)) / m;
8     center.y = (x1 * x2 * (x2 - x1) - y1 * y1 * x2 + x1 * y2 * y2) / m;
9     return center + p0;
10 }
11 void min_enclosing(vector<point<ld>> p) {
12     shuffle(p.begin(), p.end(), rngf);
13     r = 0.0;
14     point<ld> cent = p[0];
15     for (int i = 1; i < p.size(); ++i) {
16         if (abs2(cent - p[i]) <= r) continue;
17         cent = p[i], r = 0.0;
18         for (int j = 0; j < i; ++j) {
19             if (abs2(cent - p[j]) <= r) continue;
20             cent = (p[i] + p[j]) / 2, r = abs2(p[j] - cent);
21             for (int k = 0; k < j; ++k) {
22                 if (abs2(cent - p[k]) <= r) continue;
23                 cent = circenter(p[i], p[j], p[k]);
24                 r = abs2(p[k] - cent);
25             }
26         }
27     }
28 }

```

```

25     }
26 }
27 }
28 o = cent;
29 r = sqrt(r);
30 }

```

5.28 Geometry Struct 3D [4a50c9]

```

1 using ld = long double;
2
3 // 判斷數值正負 : {1:正數,0:零,-1:負數}
4 int sign(long long x) {return (x >= 0) ? ((bool)x) : -1; }
5 int sign(ld x) {return (abs(x) < 1e-9) ? 0 : (x>0 ? 1 : -1); }
6
7 template<typename T>
8 struct pt3 {
9     T x, y, z;
10    pt3(){}
11    pt3(const T &x, const T &y, const T &z):x(x),y(y),z(z){}
12    explicit operator pt3<ld>() {return pt3<ld>(x, y, z); }
13
14    pt3 operator+(pt3 b) {return {x+b.x, y+b.y, z+b.z}; }
15    pt3 operator-(pt3 b) {return {x-b.x, y-b.y, z-b.z}; }
16    pt3 operator*(T b) {return {x * b, y * b, z * b}; }
17    pt3 operator/(T b) {return {x / b, y / b, z / b}; }
18    bool operator==(pt3 b){return x==b.x&&y==b.y&&z==b.z;}
19
20    T operator*(pt3 b) {return x*b.x+y*b.y+z*b.z; }
21    pt3 operator^(pt3 b) {
22        return pt3(y * b.z - z * b.y,
23            z * b.x - x * b.z,
24            x * b.y - y * b.x);
25    }
26    friend T abs2(pt3 b) {return b * b; }
27    friend T len (pt3 b) {return sqrt(abs2(b)); }
28    friend ostream& operator<<(ostream& os, pt3 p) {
29        return os << "(" << p.x << ", " <<
30            p.y << ", " << p.z << ")";
31    }
32 };
33
34 template<typename T>
35 struct face {
36     int a, b, c; // 三角形在 vector 裡面的 index
37     pt3<T> q; // 面積向量 ( 朝外 )
38 };
39
40 /// 警告 ; v 在過程中可能被修改 · 回傳的 face 以修改後的為準
41 /// O(n^2) · 最多只有 2n-4 個面
42 /// 當凸包退化時會回傳空的凸包 · 否則回傳凸包上的每個面
43 template<typename T>
44 vector<face<T>> hull3(vector<pt3<T>> &v) {
45     int n = v.size();
46     if (n < 3) return {};
47     // don't use "==" when you use ld
48     sort(all(v), [&](pt3<T> &p, pt3<T> &q) {
49         return sign(p.x - q.x) ? (p.x < q.x) :
50             (sign(p.y - q.y) ? p.y < q.y : p.z < q.z);
51     });
52     v.resize(unique(v.begin(), v.end()) - v.begin());
53     for (int i = 2; i <= n; ++i) {
54         if (i == n) return {};
55         if (sign(len(((v[1] - v[0]) ^ (v[i] - v[0]))))) {

```

```

56         swap(v[2], v[i]);
57         break;
58     }
59 }
60 pt3<T> tmp_q = (v[1] - v[0]) ^ (v[2] - v[0]);
61 for (int i = 3; i <= n; ++i) {
62     if (i == n) return {};
63     if (sign((v[i] - v[0]) * tmp_q)) {
64         swap(v[3], v[i]);
65         break;
66     }
67 }
68
69 vector<face<T>> f;
70 vector<vector<int>> dead(n, vector<int>(n, true));
71 auto add_face = [&](int a, int b, int c) {
72     f.emplace_back(a, b, c, (v[b] - v[a]) ^ (v[c] - v[a])
73 );
74     dead[a][b] = dead[b][c] = dead[c][a] = false;
75 };
76 add_face(0, 1, 2);
77 add_face(0, 2, 1);
78
79 for (int i = 3; i < n; ++i) {
80     vector<face<T>> f2;
81     for (auto &[a, b, c, q] : f) {
82         if (sign((v[i] - v[a]) * q) > 0)
83             dead[a][b] = dead[b][c] = dead[c][a] = true;
84         else f2.emplace_back(a, b, c, q);
85     }
86     f.clear();
87     for (face<T> &F : f2) {
88         int arr[3] = {F.a, F.b, F.c};
89         for (int j = 0; j < 3; ++j) {
90             int a = arr[j], b = arr[(j + 1) % 3];
91             if (dead[b][a]) add_face(b, a, i);
92         }
93     }
94     f.insert(f.end(), all(f2));
95 }
96 return f;
97 } // 15ef50

```

5.29 Pick's Theorem

給定頂點坐標均是整點的簡單多邊形，面積 = 內部格點數 + 邊上格點數/2 - 1

6 Graph

6.1 SAT [5a6317]

```

1 struct TWO_SAT {
2     int n, N;
3     vector<vector<int>> G, rev_G;
4     deque<bool> used;
5     vector<int> order, comp;
6     deque<bool> assignment;
7     void init(int _n) {
8         n = _n;
9         N = _n * 2;
10        G.resize(N + 5);
11        rev_G.resize(N + 5);
12    }
13    void dfs1(int v) {
14        used[v] = true;
15        for (int u : G[v]) {

```

```

16            if (!used[u])
17                dfs1(u);
18        }
19        order.push_back(v);
20    }
21    void dfs2(int v, int cl) {
22        comp[v] = cl;
23        for (int u : rev_G[v]) {
24            if (comp[u] == -1)
25                dfs2(u, cl);
26        }
27    }
28    bool solve() {
29        order.clear();
30        used.assign(N, false);
31        for (int i = 0; i < N; ++i) {
32            if (!used[i])
33                dfs1(i);
34        }
35        comp.assign(N, -1);
36        for (int i = 0, j = 0; i < N; ++i) {
37            int v = order[N - i - 1];
38            if (comp[v] == -1)
39                dfs2(v, j++);
40        }
41        assignment.assign(n, false);
42        for (int i = 0; i < N; i += 2) {
43            if (comp[i] == comp[i + 1])
44                return false;
45            assignment[i / 2] = (comp[i] > comp[i + 1]);
46        }
47        return true;
48    }
49    // A or B 都是 0-based
50    void add_disjunction(int a, bool na, int b, bool nb) {
51        // na is true => ~a, na is false => a
52        // nb is true => ~b, nb is false => b
53        a = 2 * a ^ na;
54        b = 2 * b ^ nb;
55        int neg_a = a ^ 1;
56        int neg_b = b ^ 1;
57        G[neg_a].push_back(b);
58        G[neg_b].push_back(a);
59        rev_G[b].push_back(neg_a);
60        rev_G[a].push_back(neg_b);
61        return;
62    }
63    void get_result(vector<int>& res) {
64        res.clear();
65        for (int i = 0; i < n; ++i)
66            res.push_back(assignment[i]);
67    }
68 };

```

6.2 Augment Path [f8a5dd]

```

1 struct AugmentPath{
2     int n, m;
3     vector<vector<int>> G;
4     vector<int> mx, my;
5     vector<int> visx, visy;
6     int stamp;
7
8     AugmentPath(int _n, int _m) : n(_n), m(_m), G(n), mx(n,
9         -1), my(m, -1), visx(n), visy(n){

```

```

9         stamp = 0;
10    }
11
12    void add(int x, int y){
13        G[x].push_back(y);
14    }
15
16    // bb03e2
17    bool dfs1(int now){
18        visx[now] = stamp;
19
20        for (auto x : G[now]){
21            if (my[x]==-1){
22                mx[now] = x;
23                my[x] = now;
24                return true;
25            }
26        }
27        for (auto x : G[now]){
28            if (visx[my[x]]!=stamp && dfs1(my[x])){
29                mx[now] = x;
30                my[x] = now;
31                return true;
32            }
33        }
34        return false;
35    }
36
37    vector<pair<int, int>> find_max_matching(){
38        vector<pair<int, int>> ret;
39
40        while (true){
41            stamp++;
42            int tmp = 0;
43            for (int i=0 ; i<n ; i++){
44                if (mx[i]==-1 && dfs1(i)) tmp++;
45            }
46            if (tmp==0) break;
47        }
48
49        for (int i=0 ; i<n ; i++){
50            if (mx[i]!=-1){
51                ret.push_back({i, mx[i]});
52            }
53        }
54        return ret;
55    }
56
57    // 645577
58    void dfs2(int now){
59        visx[now] = true;
60
61        for (auto x : G[now]){
62            if (my[x]!=-1 && visy[x]==false){
63                visy[x] = true;
64                dfs2(my[x]);
65            }
66        }
67    }
68
69    // 要先執行 find_max_matching 一次
70    vector<pair<int, int>> find_min_vertex_cover(){
71        fill(visx.begin(), visx.end(), false);
72        fill(visy.begin(), visy.end(), false);
73    }

```

```

74     vector<pair<int, int>> ret;
75     for (int i=0 ; i<n ; i++){
76         if (mx[i]==-1) dfs2(i);
77     }
78
79     for (int i=0 ; i<n ; i++){
80         if (visx[i]==false) ret.push_back({1, i});
81     }
82     for (int i=0 ; i<m ; i++){
83         if (visy[i]==true) ret.push_back({2, i});
84     }
85
86     return ret;
87 }
88 };

```

6.3 C3C4 [875ea3]

```

1 // 0-based
2 void C3C4(vector<int> deg, vector<array<int, 2>> edges){
3     int N = deg.size();
4     int M = edges.size();
5
6     vector<int> ord(N), rk(N);
7     iota(ord.begin(), ord.end(), 0);
8     sort(ord.begin(), ord.end(), [&](int x, int y) { return
9         deg[x] > deg[y]; });
10    for (int i=0 ; i<N ; i++) rk[ord[i]] = i;
11
12    vector<vector<int>> D(N), adj(N);
13    for (auto [u, v] : edges) {
14        if (rk[u] > rk[v]) swap(u, v);
15        D[u].emplace_back(v);
16        adj[u].emplace_back(v);
17        adj[v].emplace_back(u);
18    }
19
20    vector<int> vis(N);
21
22    int c3 = 0, c4 = 0;
23    for (int x : ord) { // c3
24        for (int y : D[x]) vis[y] = 1;
25        for (int y : D[x]) for (int z : D[y]){
26            c3 += vis[z]; // xyz is C3
27        }
28        for (int y : D[x]) vis[y] = 0;
29    }
30    for (int x : ord) { // c4
31        for (int y : D[x]) for (int z : adj[y])
32            if (rk[z] > rk[x]) c4 += vis[z]++;
33        for (int y : D[x]) for (int z : adj[y])
34            if (rk[z] > rk[x]) --vis[z];
35    } // both are O(M*sqrt(M)), test @ 2022 CCPC guangzhou
36 }

```

6.4 Dinic [961b34]

```

1 // 一般圖 : O(EV^2)
2 // 二分圖 : O(EV)
3 struct Flow{
4     using T = int; // 可以換成別的类型別
5     struct Edge{
6         int v; T rc; int rid;

```

```

7     };
8     vector<vector<Edge>> G;
9     void add(int u, int v, T c){
10         G[u].push_back({v, c, G[v].size()});
11         G[v].push_back({u, 0, G[u].size()-1});
12     }
13     vector<int> dis, it;
14
15     Flow(int n){
16         G.resize(n);
17         dis.resize(n);
18         it.resize(n);
19     }
20
21     // ce56d6
22     T dfs(int u, int t, T f){
23         if (u == t || f == 0) return f;
24         for (int &i=it[u] ; i<G[u].size() ; i++){
25             auto &[v, rc, rid] = G[u][i];
26             if (dis[v]!=dis[u]+1) continue;
27             T df = dfs(v, t, min(f, rc));
28             if (df <= 0) continue;
29             rc -= df;
30             G[v][rid].rc += df;
31             return df;
32         }
33         return 0;
34     }
35
36     // e22e39
37     T flow(int s, int t){
38         T ans = 0;
39         while (true){
40             fill(dis.begin(), dis.end(), INF);
41             queue<int> q;
42             q.push(s);
43             dis[s] = 0;
44
45             while (q.size()){
46                 int u = q.front(); q.pop();
47                 for (auto [v, rc, rid] : G[u]){
48                     if (rc <= 0 || dis[v] < INF) continue;
49                     dis[v] = dis[u] + 1;
50                     q.push(v);
51                 }
52             }
53             if (dis[t]==INF) break;
54
55             fill(it.begin(), it.end(), 0);
56             while (true){
57                 T df = dfs(s, t, INF);
58                 if (df <= 0) break;
59                 ans += df;
60             }
61         }
62         return ans;
63     }
64
65     // the code below constructs minimum cut
66     void dfs_mincut(int now, vector<bool> &vis){
67         vis[now] = true;
68         for (auto &[v, rc, rid] : G[now]){
69             if (vis[v] == false && rc > 0){
70                 dfs_mincut(v, vis);
71             }
72         }
73     }

```

```

73     }
74
75     vector<pair<int, int>> construct(int n, int s, vector<
76         pair<int,int>> &E){
77         // E is G without capacity
78         vector<bool> vis(n);
79         dfs_mincut(s, vis);
80         vector<pair<int, int>> ret;
81         for (auto &[u, v] : E){
82             if (vis[u] == true && vis[v] == false){
83                 ret.emplace_back(u, v);
84             }
85         }
86         return ret;
87     };

```

6.5 Dominator Tree [52b249]

```

1 // 全部都是 0-based
2 // G 要是有向無權圖
3 // 一開始要初始化 G(N, root)
4 // 用完之後要 build
5 // G[i] = 從 root 走到 i 時，一定要走到的點且離 i 最近
6 struct DominatorTree{
7     int N;
8     vector<vector<int>> G;
9     vector<vector<int>> buckets, rg;
10    // dfn[x] = the DFS order of x
11    // rev[x] = the vertex with DFS order x
12    // par[x] = the parent of x
13    vector<int> dfn, rev, par;
14    vector<int> sdom, dom, idom;
15    vector<int> fa, val;
16    int stamp;
17    int root;
18
19    int operator [] (int x){
20        return idom[x];
21    }
22
23    DominatorTree(int _N, int _root) :
24        N(_N),
25        G(N), buckets(N), rg(N),
26        dfn(N, -1), rev(N, -1), par(N, -1),
27        sdom(N, -1), dom(N, -1), idom(N, -1),
28        fa(N, -1), val(N, -1)
29    {
30        stamp = 0;
31        root = _root;
32    }
33
34    void add_edge(int u, int v){
35        G[u].push_back(v);
36    }
37
38    void dfs(int x){
39        rev[dfn[x]] = stamp = x;
40        fa[stamp] = sdom[stamp] = val[stamp] = stamp;
41        stamp++;
42
43        for (int u : G[x]){
44            if (dfn[u]==-1){
45                dfs(u);
46                par[dfn[u]] = dfn[x];

```

```

47     }
48     rg[dfn[u]].push_back(dfn[x]);
49 }
50 }
51
52 int eval(int x, bool first){
53     if (fa[x]==x) return !first ? -1 : x;
54     int p = eval(fa[x], false);
55
56     if (p!=-1) return x;
57     if (sdom[val[x]]>sdom[val[fa[x]]]) val[x] = val[fa[x]];
58     fa[x] = p;
59
60     return !first ? p : val[x];
61 }
62
63 void link(int x, int y){
64     fa[x] = y;
65 }
66
67 void build(){
68     dfs(root);
69
70     for (int x=stamp-1 ; x>=0 ; x--){
71         for (int y : rg[x]){
72             sdom[x] = min(sdom[x], sdom[eval(y, true)]);
73         }
74         if (x>0) buckets[sdom[x]].push_back(x);
75         for (int u : buckets[x]){
76             int p = eval(u, true);
77             if (sdom[p]==x) dom[u] = x;
78             else dom[u] = p;
79         }
80         if (x>0) link(x, par[x]);
81     }
82
83     idom[root] = root;
84     for (int x=1 ; x<stamp ; x++){
85         if (sdom[x]!=dom[x]) dom[x] = dom[dom[x]];
86     }
87     for (int i=1 ; i<stamp ; i++) idom[rev[i]] = rev[dom[i]];
88 }
89 };

```

6.6 EdgeBCC [d09eb1]

```

1 // d09eb1
2 // 0-based · 支援重邊
3 struct EdgeBCC{
4     int n, m, dep, sz;
5     vector<vector<pair<int, int>>> G;
6     vector<vector<int>> bcc;
7     vector<int> dfn, low, stk, isBridge, bccId;
8     vector<pair<int, int>> edge, bridge;
9
10    EdgeBCC(int _n) : n(_n), m(0), sz(0), dfn(n), low(n), G(n), bcc(n), bccId(n) {}
11
12    void add_edge(int u, int v) {
13        edge.push_back({u, v});
14        G[u].push_back({v, m});
15        G[v].push_back({u, m++});
16    }

```

```

17    void dfs(int now, int pre) {
18        dfn[now] = low[now] = ++dep;
19        stk.push_back(now);
20
21        for (auto [x, id] : G[now]){
22            if (!dfn[x]){
23                dfs(x, id);
24                low[now] = min(low[now], low[x]);
25            }else if (id!=pre){
26                low[now] = min(low[now], dfn[x]);
27            }
28        }
29
30        if (low[now]==dfn[now]){
31            if (pre!=-1) isBridge[pre] = true;
32            int u;
33            do{
34                u = stk.back();
35                stk.pop_back();
36                bcc[sz].push_back(u);
37                bccId[u] = sz;
38            } while (u!=now);
39            sz++;
40        }
41    }
42
43    void get_bcc() {
44        isBridge.assign(m, 0);
45        dep = 0;
46        for (int i=0 ; i<n ; i++){
47            if (!dfn[i]) dfs(i, -1);
48        }
49
50        for (int i=0 ; i<m ; i++){
51            if (isBridge[i]){
52                bridge.push_back({edge[i].first , edge[i].second});
53            }
54        }
55    }
56 }
57 };

```

6.7 EnumeratePlanarFace [8df3e2]

```

1 struct PlanarGraph { // 0-based index
2     using T = int;
3     int n, m, faces;
4     vector<point<T>> v;
5     vector<vector<pair<int, int>>> D, G;
6     vector<vector<int>> F;
7     /* D : 對偶圖 · 平面圖的邊的編號是 [0, m - 1],
8        outer face 的編號是 m, face 的編號是 [m + 1, m + faces]
9        F : 存每個 inner face 的 point index */
10    vector<int> conv, nxt, vis;
11
12    PlanarGraph(int n, vector<point<T>> _v) :
13        n(n), m(0), faces(0), v(_v), G(n) {}
14
15    void add_edge(int x, int y) {
16        G[x].emplace_back(y, 2 * m);
17        G[y].emplace_back(x, 2 * m + 1);
18        conv.push_back(x);
19        conv.push_back(y);
20        m++;

```

```

21    }
22
23    vector<T> enumerate_face() {
24        D.resize(m + 1);
25        F.resize(m + 1);
26        nxt.resize(2 * m);
27        vis.resize(2 * m);
28        for (int i = 0; i < n; i++) {
29            sort(G[i].begin(), G[i].end(), [&](auto& a, auto& b) {
30                return (v[a.first]-v[i]) < (v[b.first]-v[i]);
31            }); // 極角排序用的 '<' operator
32            int sz = G[i].size(), pre = sz - 1;
33            for (int j = 0; j < sz; pre = j, j++) {
34                nxt[G[i][pre].second] = G[i][j].second ^ 1;
35            }
36        }
37
38        vector<T> ret;
39        // 這個 for 迴圈的 hash value 是 fa2180
40        for (int i = 0; i < 2 * m; i++) if (vis[i] == false){
41            vector<int> pt, outdeg;
42            for (int now = i; !vis[now]; now = nxt[now]) {
43                vis[now] = true;
44                pt.push_back(conv[now]);
45                outdeg.push_back(now / 2);
46            }
47
48            T area = -(v[pt.back()] ^ v[pt.front()]);
49            for (int j = 0; j + 1 < pt.size(); j++) {
50                area -= (v[pt[j]] ^ v[pt[j + 1]]);
51            }
52
53            if (area > 0) { // inner face
54                F.push_back(move(pt));
55                ret.push_back(area);
56                faces += 1;
57                D.emplace_back();
58                for (auto &u:outdeg) {
59                    D[m + faces].emplace_back(u, 0);
60                    D[u].emplace_back(m + faces, 1);
61                }
62            }
63            else { // outer face or a tree
64                for (auto &u:outdeg) {
65                    D[m].emplace_back(u, 0);
66                }
67            }
68        }
69        return ret;
70    }
71 }
72 };

```

6.8 Ford Fulkerson [5bc854]

```

1 const int maxn = 1e5 + 10, INF = 1e9;
2 const long long INF64 = 1e18;
3 struct edge{ int to, cap, rev; };
4 vector<edge> G[maxn];
5 int n, m, s, t, a, b, c;
6 bool vis[maxn];
7 int dfs(int v, int t, int f) {
8     // cout << v << ' ' << t << ' ' << f << '\n';
9     if (v == t) return f;
10    vis[v] = true;

```

```

11 for (edge &e: G[v]) {
12     if (!vis[e.to] && e.cap > 0) {
13         int d = dfs(e.to, t, min(f, e.cap));
14         if (d > 0) {
15             e.cap -= d, G[e.to][e.rev].cap += d;
16             return d;
17         }
18     }
19 }
20 return 0;
21 }
22 int ford_fulkerson(int s, int t) {
23     int flow = 0, f;
24     // for (int i = 0; i < n; i++) {
25     //     cout << i << " : ";
26     //     for (edge e: G[i])
27     //         cout << '(' << e.to << ', ' << e.cap << ')' <<
28     //             ' ';
29     //     cout << '\n';
30     // }
31     do {
32         memset(vis, false, sizeof(vis));
33         f = dfs(s, t, INF);
34         // for (int i = 0; i < n; i++) {
35         //     cout << i << " : ";
36         //     for (edge e: G[i])
37         //         cout << '(' << e.to << ', ' << e.cap << ')' <<
38         //             ' ';
39         //     cout << '\n';
40         // }
41         // cout << f << '\n';
42         flow += f;
43     } while (f > 0);
44     return flow;
45 }
46 void init(int n) {
47     for (int i = 0; i < n; i++) G[i].clear();
48 }
49 int main() {
50     cin >> n >> m >> s >> t;
51     init(n);
52     while (m--) {
53         cin >> a >> b >> c;
54         G[a].push_back((edge){b, c, (int)G[b].size()});
55         G[b].push_back((edge){a, 0, (int)G[a].size() - 1});
56     }
57     cout << ford_fulkerson(s, t) << '\n';
58     return 0;
59 }

```

6.9 HLD [f57ec6]

```

1 #include <bits/stdc++.h>
2 #define int long long
3 using namespace std;
4
5 const int N = 100005;
6 vector<int> G[N];
7 struct HLD {
8     vector<int> pa, sz, depth, mxson, topf, id;
9     int n, idcnt = 0;
10    HLD(int _n) : n(_n), pa(_n + 1), sz(_n + 1), depth(_n +
11        1), mxson(_n + 1), topf(_n + 1), id(_n + 1) {}
12    void dfs1(int v = 1, int p = -1) {
13        pa[v] = p; sz[v] = 1; mxson[v] = 0;

```

```

13        depth[v] = (p == -1 ? 0 : depth[p] + 1);
14        for (int u : G[v]) {
15            if (u == p) continue;
16            dfs1(u, v);
17            sz[v] += sz[u];
18            if (sz[u] > sz[mxson[v]]) mxson[v] = u;
19        }
20    }
21    void dfs2(int v = 1, int top = 1) {
22        id[v] = ++idcnt;
23        topf[v] = top;
24        if (mxson[v]) dfs2(mxson[v], top);
25        for (int u : G[v]) {
26            if (u == mxson[v] || u == pa[v]) continue;
27            dfs2(u, u);
28        }
29    }
30    // query 為區間資料結構
31    int path_query(int a, int b) {
32        int res = 0;
33        while (topf[a] != topf[b]) { /// 若不在同一條鍊上
34            if (depth[topf[a]] < depth[topf[b]]) swap(a, b);
35            res = max(res, 0ll); // query : l = id[topf[a]],
36                r = id[a]
37            a = pa[topf[a]];
38        }
39        /// 此時已在同一條鍊上
40        if (depth[a] < depth[b]) swap(a, b);
41        res = max(res, 0ll); // query : l = id[b], r = id[a]
42        return res;
43    }
44 }

```

6.10 Kosaraju SCC [c7d5aa]

```

1 // 給定一個有向圖，迴傳傳縮點後的圖、SCC 的資訊
2 // 所有點都以 based-0 編號
3 // 函式：
4 // .compress:  $O(n \log n)$  計算  $G3$ 、 $SCC$ 、 $SCC\_id$  的資訊，並把縮
5 // 點後的結果存在 result 裡
6 //  $SCC[i]$  = 某個  $SCC$  中的所有點
7 //  $SCC\_id[i]$  = 第  $i$  個點在第幾個  $SCC$ 
8 struct SCC_compress {
9     int N, M, sz;
10    vector<vector<int>> G, inv_G, result;
11    vector<pair<int, int>> edges;
12    vector<bool> vis;
13    vector<int> order;
14
15    vector<vector<int>> SCC;
16    vector<int> SCC_id;
17
18    SCC_compress(int _N) :
19        N(_N), M(0), sz(0),
20        G(N), inv_G(N),
21        vis(N), SCC_id(N) {}
22
23    vector<int> operator [] (int x){
24        return result[x];
25    }
26
27    void add_edge(int u, int v){
28        G[u].push_back(v);

```

```

29        inv_G[v].push_back(u);
30        edges.push_back({u, v});
31        M++;
32    }
33
34    void dfs1(vector<vector<int>> &G, int now){
35        vis[now] = 1;
36        for (auto x : G[now]) if (!vis[x]) dfs1(G, x);
37        order.push_back(now);
38    }
39
40    void dfs2(vector<vector<int>> &G, int now){
41        SCC_id[now] = SCC.size() - 1;
42        SCC.back().push_back(now);
43        vis[now] = 1;
44        for (auto x : G[now]) if (!vis[x]) dfs2(G, x);
45    }
46
47    void compress(){
48        fill(vis.begin(), vis.end(), 0);
49        for (int i=0; i<N; i++) if (!vis[i]) dfs1(G, i);
50
51        fill(vis.begin(), vis.end(), 0);
52        reverse(order.begin(), order.end());
53        for (int i=0; i<N; i++){
54            if (!vis[order[i]]){
55                SCC.push_back(vector<int>());
56                dfs2(inv_G, order[i]);
57            }
58        }
59
60        result.resize(SCC.size());
61        sz = SCC.size();
62        for (auto [u, v] : edges){
63            if (SCC_id[u] != SCC_id[v]) result[SCC_id[u]].
64                push_back(SCC_id[v]);
65        }
66        for (int i=0; i<SCC.size(); i++){
67            sort(result[i].begin(), result[i].end());
68            result[i].resize(unique(result[i].begin(), result
69                [i].end())-result[i].begin());
70        }
71    }
72 }

```

6.11 Kuhn Munkres [e66c35]

```

1 //  $O(n^3)$  找到最大權匹配
2 struct KuhnMunkres {
3     int n; // max(n, m)
4     vector<vector<int>> G;
5     vector<int> match, lx, ly, visx, visy;
6     vector<int> slack;
7     int stamp = 0;
8
9     KuhnMunkres(int n) : n(n), G(n, vector<int>(n)), lx(n),
10        ly(n), slack(n), match(n), visx(n), visy(n) {}
11
12    void add(int x, int y, int w){
13        G[x][y] = max(G[x][y], w);
14    }
15
16    bool dfs(int i, bool aug){ // aug = true 表示要更新 match
17        if (visx[i] == stamp) return false;
18        visx[i] = stamp;

```

```

18     for (int j=0 ; j<n ; j++){
19         if (visy[j]==stamp) continue;
20         int d = lx[i]+ly[j]-G[i][j];
21
22         if (d==0){
23             visy[j] = stamp;
24             if (match[j]==-1 || dfs(match[j], aug)){
25                 if (aug){
26                     match[j] = i;
27                 }
28                 return true;
29             }
30         }
31         else{
32             slack[j] = min(slack[j], d);
33         }
34     }
35     return false;
36 }
37
38 bool augment(){
39     for (int j=0 ; j<n ; j++){
40         if (visy[j]!=stamp && slack[j]==0){
41             visy[j] = stamp;
42             if (match[j]==-1 || dfs(match[j], false)){
43                 return true;
44             }
45         }
46     }
47     return false;
48 }
49
50 void relabel(){
51     int delta = INF;
52     for (int j=0 ; j<n ; j++){
53         if (visy[j]!=stamp) delta = min(delta, slack[j]);
54     }
55     for (int i=0 ; i<n ; i++){
56         if (visx[i]==stamp) lx[i] -= delta;
57     }
58     for (int j=0 ; j<n ; j++){
59         if (visy[j]==stamp) ly[j] += delta;
60         else slack[j] -= delta;
61     }
62 }
63
64 int solve(){
65
66     for (int i=0 ; i<n ; i++){
67         lx[i] = 0;
68         for (int j=0 ; j<n ; j++){
69             lx[i] = max(lx[i], G[i][j]);
70         }
71     }
72
73     fill(ly.begin(), ly.end(), 0);
74     fill(match.begin(), match.end(), -1);
75
76     for(int i = 0; i < n; i++) {
77         fill(slack.begin(), slack.end(), INF);
78         stamp++;
79         if(dfs(i, true)) continue;
80
81         while(augment()==false) relabel();
82         stamp++;
83         dfs(i, true);

```

```

84     }
85
86     int ans = 0;
87     for (int j=0 ; j<n ; j++){
88         if (match[j]!=-1){
89             ans += G[match[j]][j];
90         }
91     }
92     return ans;
93 }
94 };

```

6.12 LCA [5b6a5b]

```

1 // 1-based · 可以支援森林 · 0 是超級源點 · 所有樹都要跟他建邊
2 struct Tree{
3     int N, M = 0, H;
4     vector<int> parent, dep;
5     vector<vector<int>>> G, LCA;
6
7     Tree(int _N) : N(_N+1), H(__lg(N)+1), parent(N, -1), dep(N, 0){
8         LCA.resize(H, vector<int>(N, 0));
9     }
10
11     void add_edge(int u, int v){
12         M++;
13         G[u].push_back(v);
14         G[v].push_back(u);
15     }
16
17     void dfs(int now = 0, int pre = 0){
18         dep[now] = dep[pre]+1;
19         parent[now] = pre;
20         for (auto x : G[now]){
21             if (x==pre) continue;
22             dfs(x, now);
23         }
24     }
25
26     void build_LCA(int root = 0){
27         dfs();
28         for (int i=0 ; i<N ; i++) LCA[0][i] = parent[i];
29         for (int i=1 ; i<H ; i++){
30             for (int j=0 ; j<N ; j++){
31                 LCA[i][j] = LCA[i-1][LCA[i-1][j]];
32             }
33         }
34     }
35
36     int jump(int u, int step){
37         for (int i=0 ; i<H ; i++){
38             if (step&(1<<i)) u = LCA[i][u];
39         }
40         return u;
41     }
42
43     int get_LCA(int u, int v){
44         if (dep[u]<dep[v]) swap(u, v);
45         u = jump(u, dep[u]-dep[v]);
46         if (u==v) return u;
47         for (int i=H-1 ; i>=0 ; i--){
48             if (LCA[i][u]!=LCA[i][v]){
49                 u = LCA[i][u];
50                 v = LCA[i][v];

```

```

51     }
52 }
53     return parent[u];
54 }
55 };

```

6.13 MCMF [0d5244]

```

1 struct Flow {
2     struct Edge {
3         int u, rc, k, rv;
4     };
5
6     vector<vector<Edge>>> G;
7     vector<int> par, par_eid;
8     Flow(int n) : G(n+1), par(n+1), par_eid(n+1) {}
9
10    // v->u, capacity: c, cost: k
11    void add(int v, int u, int c, int k){
12        G[v].push_back({u, c, k, G[u].size()});
13        G[u].push_back({v, 0, -k, G[v].size()-1});
14    }
15
16    // 6d1140
17    int spfa(int s, int t){
18        fill(par.begin(), par.end(), -1);
19        vector<int> dis(par.size(), INF);
20        vector<bool> in_q(par.size(), false);
21        queue<int> Q;
22        dis[s] = 0;
23        in_q[s] = true;
24        Q.push(s);
25
26        while (!Q.empty()){
27            int v = Q.front();
28            Q.pop();
29            in_q[v] = false;
30
31            for (int i=0 ; i<G[v].size() ; i++){
32                auto [u, rc, k, rv] = G[v][i];
33                if (rc>0 && dis[v]+k<dis[u]){
34                    dis[u] = dis[v]+k;
35                    par[u] = v;
36                    par_eid[u] = i;
37                    if (!in_q[u]) Q.push(u);
38                    in_q[u] = true;
39                }
40            }
41        }
42
43        return dis[t];
44    }
45
46    // return <max flow, min cost>, d7e7ad
47    pair<int, int> flow(int s, int t){
48        int fl = 0, cost = 0, d;
49        while ((d = spfa(s, t))<INF){
50            int cur = INF;
51            for (int v=t ; v!=s ; v=par[v]){
52                cur = min(cur, G[par[v]][par_eid[v]].rc);
53            }
54            fl += cur;
55            cost += d*cur;
56            for (int v=t ; v!=s ; v=par[v]){
57                G[par[v]][par_eid[v]].rc -= cur;
58                G[v][G[par[v]][par_eid[v]].rv].rc += cur;

```

```

58     }
59 }
60 return {f1, cost};
61 }
62 };

```

6.14 Tarjan SCC [8b2350]

```

1 struct tarjan_SCC {
2     int now_T, now_SCCs;
3     vector<int> dfn, low, SCC;
4     stack<int> S;
5     vector<vector<int>> E;
6     vector<bool> vis, in_stack;
7
8     tarjan_SCC(int n) {
9         init(n);
10    }
11    void init(int n) {
12        now_T = now_SCCs = 0;
13        dfn = low = SCC = vector<int>(n);
14        E = vector<vector<int>>(n);
15        S = stack<int>();
16        vis = in_stack = vector<bool>(n);
17    }
18    void add(int u, int v) {
19        E[u].push_back(v);
20    }
21    void build() {
22        for (int i = 0; i < dfn.size(); ++i) {
23            if (!dfn[i]) dfs(i);
24        }
25    }
26    void dfs(int v) {
27        now_T++;
28        vis[v] = in_stack[v] = true;
29        dfn[v] = low[v] = now_T;
30        S.push(v);
31        for (auto &i:E[v]) {
32            if (!vis[i]) {
33                vis[i] = true;
34                dfs(i);
35                low[v] = min(low[v], low[i]);
36            }
37            else if (in_stack[i]) {
38                low[v] = min(low[v], dfn[i]);
39            }
40        }
41        if (low[v] == dfn[v]) {
42            int tmp;
43            do {
44                tmp = S.top();
45                S.pop();
46                SCC[tmp] = now_SCCs;
47                in_stack[tmp] = false;
48            } while (tmp != v);
49            now_SCCs += 1;
50        }
51    }
52 };

```

6.15 Theorem

- 任意圖
 - 最大匹配 + 最小邊覆蓋 = n (不能有孤點)

- 點覆蓋的補集是獨立集。最小點覆蓋 + 最大獨立集 = n
- w (最小權點覆蓋) + w (最大權獨立集) = $\sum w_v$
- (帶點權的二分圖可以用最小割解。構造請參考 Augment Path.cpp)

• 二分圖

- 最小點覆蓋 = 最大匹配 = n - 最大獨立集

• 只有邊帶權的二分圖

- w-vertex-cover (帶權點覆蓋): 每條邊的兩個連接點被選中的次數總和至少要是 w_e 。
- w-weight matching (帶權匹配)
- minimum vertex count of w-vertex-cover = maximum weight count of w-weight matching (一個點可以被選很多次, 但邊不行)

• 點、邊都帶權的二分圖的定理

- b-matching: 假設 v 的點權是 b_v 。那所有 v 的匹配邊 e 的權重都要滿足 $\sum w_e \leq b_v$ 。
- The maximum w-weight of a b-matching equals the minimum b-weight of vertices in a w-vertex-cover.

6.16 Tree Isomorphism [6f5c9e]

```

1 // Please change "mul" to a random number (> N) before submit
2 struct Tree_Hash { // 1-based
3     using T = __int128;
4     const T mod = (T)(1e18) + 3, mul = 435216;
5     vector<vector<int>> E;
6     Tree_Hash(int _n) : E(_n + 1) {}
7     void add_edge(int u, int v) {
8         E[u].push_back(v); E[v].push_back(u);
9     }
10    pair<T, T> h(int v, int p = -1) {
11        vector<pair<T, T>> s;
12        for (auto &i:E[v]) if (i != p) {
13            s.push_back(h(i, v));
14        }
15        T res = s.size() + 1, prod = mul;
16        sort(s.begin(), s.end());
17        for (auto &[i, j] : s) {
18            res = (res + i * prod) % mod;
19            prod = prod * j % mod;
20        }
21        return {res, prod};
22    }
23    T cal(int v = 1) {
24        int n = 0, c1 = -1, c2 = -1;
25        auto dfs = [&](auto&& f, int v, int p) -> int {
26            int sz = 1, mx = 0;
27            for (int u : E[v]) if (u != p) {
28                int s = f(f, u, v);
29                sz += s;
30                mx = max(mx, s);
31            }
32            mx = max(mx, n - sz);
33            if (n && mx * 2 <= n) (~c1 ? c2 : c1) = v;
34            return sz;
35        };
36        n = dfs(dfs, v, -1); dfs(dfs, v, -1);
37        E.push_back({c1, (c2 == -1) ? c1 : c2});
38        return h(E.size() - 1).first;
39    }
40 };

```

6.17 prufer reverse [2f831d]

```

1 // K_n 有  $n \wedge (n - 2)$  種生成樹
2 vector<pair<int, int>> prufer_reverse(vector<int> &v) {
3     int n = v.size() + 2, mn = 1, mx = n; // 1-based
4     vector<int> deg(mx + 1);
5     for (auto &i:v) deg[i] += 1;
6     int ind = mn;
7     while (deg[ind] != 0) ind += 1;
8     int leaf = ind;
9     vector<pair<int, int>> res;
10    for (auto &i:v) {
11        res.emplace_back(leaf, i);
12        deg[i] -= 1;
13        if (deg[i] == 0 && i < ind) leaf = i;
14        else {
15            do ind += 1; while (deg[ind] != 0);
16            leaf = ind;
17        }
18    }
19    res.emplace_back(leaf, mx);
20    return res;
21 }

```

6.18 圓方樹 [5a6c9f]

```

1 struct BCC_AP { // 0-based, remember to build, 不支援重邊
2     int n, nbcc; // 注意孤點不會有任何方點連接
3     vector<vector<int>> E, F; // id >= n: 方點
4     vector<int> pa, dep, low, stk, paf, depf;
5     void dfs(int v, int p) {
6         dep[v] = low[v] = ~p ? dep[p] + 1 : 0;
7         stk.push_back(v), pa[v] = p;
8         for (auto& u : E[v]) if (u != p) {
9             if (low[u] == -1) {
10                dfs(u, v), low[v] = min(low[v], low[u]);
11                if (low[u] >= dep[v]) {
12                    int id = nbcc++, x;
13                    do {
14                        x = stk.back(), stk.pop_back();
15                        F[id + n].push_back(x), F[x].push_back(id + n);
16                    } while (x != u);
17                    F[id + n].push_back(v), F[v].push_back(id + n);
18                }
19                } else low[v] = min(low[v], dep[u]);
20            }
21        }
22    bool is_bridge(int u, int v) {
23        return (F[bcc_id(u, v) + n].size() == 2);
24    }
25    // 判斷原圖上的一個點是不是割點
26    bool is_cut(int x) { return F[x].size() != 1; }
27    // 回傳第 id 個 bcc 的每個點 (0-based)
28    vector<int> bcc(int id) { return F[id + n]; }
29    // 對於邊 (u, v) 回傳包含它的 bcc id (每條邊都恰好被一個 bcc 包含)
30    int bcc_id(int u, int v) { // starts from 0
31        return paf[depf[u] < depf[v] ? v : u] - n;
32    }
33    // 計算在新圖上的 depf, paf 陣列。呼叫 bcc_id() 的時候會用到
34    void dfs2(int v, int p) {
35        depf[v] = ~p ? depf[p] + 1 : 0, paf[v] = p;
36    }

```

```

34     for (int u : F[v]) if (u != p) dfs2(u, v);
35 }
36 void build() {
37     low.assign(n, -1);
38     for (int i = 0; i < n; ++i) if (low[i] == -1)
39         dfs(i, -1), dfs2(i, -1);
40 }
41 void add_edge(int u, int v) {
42     E[u].push_back(v), E[v].push_back(u);
43 }
44 BCC_AP (int _n) : n(_n), nbcc(0), E(n), F(2 * n), pa(n),
45     dep(n), low(n), stk(), paf(n * 2), depf(n * 2) {}

```

6.19 最大權閉合圖 [6ca663]

```

1 // 邊  $u \rightarrow v$  表示選  $u$  就要選  $v$  ( $\theta$ -based)
2 // 保證回傳值非負
3 // 構造：呼叫 construct · 選那些 vis 是 true 的點。
4 // 一般圖： $O(n^2m)$  / 二分圖： $O(m\sqrt{n})$ 
5 template<typename U>
6 U maximum_closure(vector<U> w, vector<pair<int,int>> EV) {
7     int n = w.size(), S = n + 1, T = n + 2;
8     Flow G(T + 5); // Graph/Dinic.cpp
9     U sum = 0;
10    for (int i = 0; i < n; ++i) {
11        if (w[i] > 0) {
12            G.add(S, i, w[i]);
13            sum += w[i];
14        }
15        else if (w[i] < 0) {
16            G.add(i, T, abs(w[i]));
17        }
18    }
19    for (auto &[u, v] : EV) { // 請務必確保  $INF > \sum w_i$ 
20        G.add(u, v, INF);
21    }
22    U cut = G.flow(S, T);
23    return sum - cut;
24 }

```

7 Math

7.1 Burnside's Lemma

$$\sum_{k=1}^n \frac{c(k)}{n}$$

- n ：有多少種置換方式（例如：旋轉方式）
- $c(k)$ ：所有可能中，經過 k 次旋轉後，仍不會和別人相同的方式的數量

7.2 CRT [6a6da6]

```

1 using ll = __int128;
2 //  $a * p.first + b * p.second = \gcd(a, b)$ 
3 pair<ll, ll> extgcd(ll a, ll b) {
4     if (b == 0) return {1, 0};
5     auto [y, x] = extgcd(b, a % b);
6     return pair<ll, ll>(x, y - (a / b) * x);
7 }
8
9 pair<ll, ll> CRT(ll x1, ll m1, ll x2, ll m2) {

```

```

10     ll g = __gcd(m1, m2);
11     if ((x2 - x1) % g) return make_pair(-1, -1); // no sol
12     m1 /= g, m2 /= g;
13     pair<ll, ll> p = extgcd(m1, m2);
14     ll lcm = m1 * m2 * g;
15     ll res = p.first * (x2 - x1) * m1 + x1;
16     // be careful with overflow
17     return make_pair((res % lcm + lcm) % lcm, lcm);
18 }
19
20 pair<ll, ll> CRT_arr(vector<ll> &a, vector<ll> &m) {
21     pair<ll, ll> res(a[0], m[0]);
22     for (int i = 1; i < a.size(); ++i) {
23         res = CRT(res.first, res.second, a[i], m[i]);
24         if (res.first == -1) break;
25     }
26     return res;
27 }

```

7.3 Catalan Number

任意括號序列： $C_n = \frac{1}{n+1} \binom{2n}{n}$

7.4 Floor Sum [5d1c9e]

```

1 // Returns  $\sum_{i=0}^{n-1} \text{floor}((a*i + b) / m)$ 
2 //  $\log(\max(a, m))$ 
3 int floor_sum(int n, int m, int a, int b) {
4     int ans = 0;
5     while (true) {
6         if (a >= m) {
7             ans += (n - 1) * n * (a / m) / 2;
8             a %= m;
9         }
10        if (b >= m) {
11            ans += n * (b / m);
12            b %= m;
13        }
14        int y_max = a * n + b;
15        if (y_max < m) break;
16        n = y_max / m;
17        b = y_max % m;
18        swap(m, a);
19    }
20    return ans;
21 }

```

7.5 Josephus Problem [e0ed50]

```

1 // 有  $n$  個人，第偶數個報數的人被刪掉，問第  $k$  個被踢掉的是誰
2 int solve(int n, int k){
3     if (n==1) return 1;
4     if (k<=(n+1)/2){
5         if (2*k>n) return 2*k%n;
6         else return 2*k;
7     }else{
8         int res=solve(n/2, k-(n+1)/2);
9         if (n&1) return 2*res+1;
10        else return 2*res-1;
11    }
12 }

```

7.6 Lagrange any x [1f2c26]

```

1 // init: ( $x_1, y_1$ ), ( $x_2, y_2$ ) in a vector
2 struct Lagrange{
3     int n;
4     vector<pair<int, int>> v;
5
6     Lagrange(vector<pair<int, int>> &_v){
7         n = _v.size();
8         v = _v;
9     }
10
11    //  $O(n^2 \log \text{MAX}_A)$ 
12    int solve(int x){
13        int ret = 0;
14        for (int i=0; i<n; i++){
15            int now = v[i].second;
16            for (int j=0; j<n; j++){
17                if (i==j) continue;
18                now *= ((x-v[j].first+MOD)%MOD);
19                now %= MOD;
20                now *= (qp((v[i].first-v[j].first+MOD)%MOD,
21                    MOD-2)+MOD)%MOD;
22                now %= MOD;
23            }
24            ret = (ret+now)%MOD;
25        }
26        return ret;
27    }
28 };

```

7.7 Lagrange continuous x [57536a]

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 const int MAX_N = 5e5 + 10;
5 const int mod = 1e9 + 7;
6
7 long long inv_fac[MAX_N];
8
9 inline int fp(long long x, int y) {
10     int ret = 1;
11     for (; y; y >>= 1) {
12         ret = (y & 1) ? (ret * x % mod) : ret;
13         x = x * x % mod;
14     }
15     return ret;
16 }
17
18 // TO USE THIS TEMPLATE, YOU MUST MAKE SURE THAT THE MOD
19 // NUMBER IS A PRIME.
20 struct Lagrange {
21     /*
22      * Initialize a polynomial with  $f(x_0)$ ,  $f(x_0 + 1)$ , ...,  $f(x_0 + n)$ .
23      * This determines a polynomial  $f(x)$  whose degree is at most  $n$ .
24      * Then you can call  $\text{sample}(x)$  and you get the value of  $f(x)$ .
25      * Complexity of  $\text{init}()$  and  $\text{sample}()$  are both  $O(n)$ .
26      */
27     int m, shift; //  $m = n + 1$ 
28     vector<int> v, mul;
29     // You can use this function if you don't have inv_fac array already.

```

```

29 void construct_inv_fac() {
30     long long fac = 1;
31     for (int i = 2; i < MAX_N; ++i) {
32         fac = fac * i % mod;
33     }
34     inv_fac[MAX_N - 1] = fp(fac, mod - 2);
35     for (int i = MAX_N - 1; i >= 1; --i) {
36         inv_fac[i - 1] = inv_fac[i] * i % mod;
37     }
38 }
39 // You call init() many times without having a second
    instance of this struct.
40 void init(int X_0, vector<int> &u) {
41     v = u;
42     shift = ((1 - X_0) % mod + mod) % mod;
43     if (v.size() == 1) v.push_back(v[0]);
44     m = v.size();
45     mul.resize(m);
46 }
47 // You can use sample(x) instead of sample(x % mod).
48 int sample(int x) {
49     x = ((long long)x + shift) % mod;
50     x = (x < 0) ? (x + mod) : x;
51     long long now = 1;
52     for (int i = m; i >= 1; --i) {
53         mul[i - 1] = now;
54         now = now * (x - i) % mod;
55     }
56     int ret = 0;
57     bool neg = (m - 1) & 1;
58     now = 1;
59     for (int i = 1; i <= m; ++i) {
60         int up = now * mul[i - 1] % mod;
61         int down = inv_fac[m - i] * inv_fac[i - 1] % mod;
62         int tmp = ((long long)v[i - 1] * up % mod) * down
63             % mod;
64         ret += (neg && tmp) ? (mod - tmp) : (tmp);
65         ret = (ret >= mod) ? (ret - mod) : ret;
66         now = now * (x - i) % mod;
67         neg ^= 1;
68     }
69     return ret;
70 };
71
72 int main() {
73     int n; cin >> n;
74     vector<int> v(n);
75     for (int i = 0; i < n; ++i) {
76         cin >> v[i];
77     }
78     Lagrange L;
79     L.construct_inv_fac();
80     L.init(0, v);
81     int x; cin >> x;
82     cout << L.sample(x);
83 }

```

7.8 Linear Mod Inverse [ecf71e]

```

1 // 線性求 1-based a[i] 對 p 的乘法反元素
2 vector<int> s(n+1, 1), invS(n+1), invA(n+1);
3 for (int i=1 ; i<=n ; i++) s[i] = s[i-1]*a[i]%p;
4 invS[n] = qp(s[n], p-2, p);
5 for (int i=n ; i>=1 ; i--) invS[i-1] = invS[i]*a[i]%p;

```

```

6 for (int i=1 ; i<=n ; i++) invA[i] = invS[i]*s[i-1]%p;

```

7.9 Lucas's Theorem [b37dcf]

```

1 // 對於很大的  $C^n_n\{m\}$  對質數  $p$  取模。只要  $p$  不大就可以用。
2 int Lucas(int n, int m, int p){
3     if (m==0) return 1;
4     return (C(n%p, m%p, p)*Lucas(n/p, m/p, p)%p);
5 }

```

7.10 Matrix [8d1a23]

```

1 struct Matrix{
2     int n, m;
3     vector<vector<int>> arr;
4
5     Matrix(int _n, int _m){
6         n = _n;
7         m = _m;
8         arr.assign(n, vector<int>(m));
9     }
10
11     vector<int> & operator [] (int i){
12         return arr[i];
13     }
14
15     Matrix operator * (Matrix b){
16         Matrix ret(n, b.m);
17         for (int i=0 ; i<n ; i++){
18             for (int j=0 ; j<b.m ; j++){
19                 for (int k=0 ; k<m ; k++){
20                     ret.arr[i][j] += arr[i][k]*b.arr[k][j]%
21                         MOD;
22                     ret.arr[i][j] %= MOD;
23                 }
24             }
25         }
26         return ret;
27     }
28
29     Matrix pow(int p){
30         Matrix ret(n, n), mul = *this;
31         for (int i=0 ; i<n ; i++){
32             ret.arr[i][i] = 1;
33         }
34
35         for ( ; p ; p>>=1){
36             if (p&1) ret = ret*mul;
37             mul = mul*mul;
38         }
39
40         return ret;
41     }
42
43     int det(){
44         vector<vector<int>> arr = this->arr;
45         bool flag = false;
46         for (int i=0 ; i<n ; i++){
47             int target = -1;
48             for (int j=i ; j<n ; j++){
49                 if (arr[j][i]){
50                     target = j;
51                     break;
52                 }

```

```

53         }
54         if (target==-1) return 0;
55         if (i!=target){
56             swap(arr[i], arr[target]);
57             flag = !flag;
58         }
59
60         for (int j=i+1 ; j<n ; j++){
61             if (!arr[j][i]) continue;
62             int freq = arr[j][i]*qp(arr[i][i], MOD-2)%MOD
63                 ;
64             for (int k=i ; k<n ; k++){
65                 arr[j][k] -= freq*arr[i][k];
66                 arr[j][k] = (arr[j][k]%MOD+MOD)%MOD;
67             }
68         }
69
70         int ret = !flag ? 1 : MOD-1;
71         for (int i=0 ; i<n ; i++){
72             ret *= arr[i][i];
73             ret %= MOD;
74         }
75         return ret;
76     }
77 };

```

7.11 Matrix 01 [8d542a]

```

1 const int MAX_N = (1LL<<12);
2 struct Matrix{
3     int n, m;
4     vector<bitset<MAX_N>> arr;
5
6     Matrix(int _n, int _m){
7         n = _n;
8         m = _m;
9         arr.resize(n);
10    }
11
12    Matrix operator * (Matrix b){
13        Matrix b_t(b.m, b.n);
14        for (int i=0 ; i<b.n ; i++){
15            for (int j=0 ; j<b.m ; j++){
16                b_t.arr[j][i] = b.arr[i][j];
17            }
18        }
19
20        Matrix ret(n, b.m);
21        for (int i=0 ; i<n ; i++){
22            for (int j=0 ; j<b.m ; j++){
23                ret.arr[i][j] = ((arr[i]&b_t.arr[j]).count()
24                    &1);
25            }
26        }
27        return ret;
28    }
};

```

7.12 Matrix Tree Theorem

目標：給定一張無向圖。問他的生成樹數量。
方法：先把所有自環刪掉。定義 Q 為以下矩陣

$$Q_{i,j} = \begin{cases} \deg(v_i) & \text{if } i = j \\ -(邊v_i v_j \text{ 的數量}) & \text{otherwise} \end{cases}$$

接著刪掉 Q 的第一個 row 跟 column，它的 determinant 就是答案。

目標：給定一張有向圖，問他的以 r 為根，可以走到所有點生成樹數量。

方法：先把所有自環刪掉，定義 Q 為以下矩陣

$$Q_{i,j} = \begin{cases} \deg_{in}(v_i) & \text{if } i = j \\ -(邊v_i v_j \text{ 的數量}) & \text{otherwise} \end{cases}$$

接著刪掉 Q 的第 r 個 row 跟 column，它的 determinant 就是答案。

7.13 Miller Rabin [5694bf]

```
1 // O(k log^3 n), k = llsprp.size()
2 using Uint = unsigned long long;
3 Uint modmul(Uint a, Uint b, Uint m) {
4     int ret = a*b - m*(Uint)((long double)a*b/m);
5     return ret + m*(ret < 0) - m*(ret >= (int)m);
6 }
7
8 int qp(int b, int p, int m){
9     int ret = 1;
10    for (; p; p>>=1){
11        if (p&1) ret = modmul(ret, b, m);
12        b = modmul(b, b, m);
13    }
14    return ret;
15 }
16
17 // ed23aa
18 vector<int> llsprp = {2, 325, 9375, 28178, 450775, 9780504,
19                     1795265022};
20 bool is_prime(int n, vector<int> sprp = llsprp){
21     if (n==2) return 1;
22     if (n<2 || n%2==0) return 0;
23
24     int t = 0;
25     int u = n-1;
26     for (; u%2==0; t++) u>>=1;
27
28     for (int i=0; i<sprp.size(); i++){
29         int a = sprp[i]%n;
30         if (a==0 || a==1 || a==n-1) continue;
31         int x = qp(a, u, n);
32         if (x==1 || x==n-1) continue;
33         for (int j=0; j<t; j++){
34             x = modmul(x, x, n);
35             if (x==1) return 0;
36             if (x==n-1) break;
37         }
38         if (x==n-1) continue;
39         return false;
40     }
41
42     return true;
43 }
```

7.14 Pollard Rho [a5daef]

```
1 mt19937 seed(chrono::steady_clock::now().time_since_epoch().
2   count());
3 int rnd(int l, int r){
4     return uniform_int_distribution<int>(l, r)(seed);
5 }
```

```
5 // O(n^{1/4}) 回傳 1 或自己的因數、記得先判斷 n 是不是質數
6   (用 Miller-Rabin)
7 // c1670c
8 int Pollard_Rho(int n){
9     int s = 0, t = 0;
10    int c = rnd(1, n-1);
11
12    int step = 0, goal = 1;
13    int val = 1;
14
15    for (goal=1; ; goal<=1, s=t, val=1){
16        for (step=1; step<=goal; step++){
17
18            t = ((__int128)t*t+c)%n;
19            val = ((__int128)val*abs(t-s)%n;
20
21            if ((step % 127) == 0){
22                int d = __gcd(val, n);
23                if (d>1) return d;
24            }
25        }
26
27        int d = __gcd(val, n);
28        if (d>1) return d;
29    }
30 }
```

7.15 Stirling's formula

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

7.16 Theorem

- $1 \sim x$ 質數的數量 $\approx \frac{x}{\ln x}$
- x 的因數的數量 $\approx x^{\frac{1}{3}}$
- x 的質因數的數量 $\approx \log \log x$
- p is a prime number $\Leftrightarrow (p-1)! \equiv -1 \pmod{p}$
- 每個正整數都可以表示成四個整數的平方和
- 任何大於 2 的整數都可以表示成兩個質數的和
- $n^{k-2} \cdot \prod_{i=1}^k s_i$ 個點、 k 的連通塊，加上 $k-1$ 條邊使得變成一個連通圖的方法數，其中每個連通塊有 s_i 個點

7.17 josephus [0be067]

```
1 // n 個人，每 k 個人就刪除的約瑟夫遊戲
2 int josephus(int n, int k) {
3     if (n == 1)
4         return 0;
5     if (k == 1)
6         return n-1;
7     if (k > n)
8         return (josephus(n-1, k) + k) % n;
9     int cnt = n / k;
10    int res = josephus(n - cnt, k);
11    res -= n % k;
12    if (res < 0)
13        res += n;
14    else
```

```
15     res += res / (k - 1);
16     return res;
17 }
```

7.18 二元一次方程式

$$\begin{cases} ax + by = e \\ cx + dy = f \end{cases} = \begin{cases} x = \frac{ed-bf}{ad-bc} \\ y = \frac{af-ec}{ad-bc} \end{cases}$$

若 $x = \frac{a}{0}$ 且 $y = \frac{b}{0}$ ，則代表無限多組解。若 $x = \frac{a}{0}$ 且 $y = \frac{b}{0}$ ，則代表無解。

7.19 數論分塊 [8ccab5]

```
1 // 時間複雜度為 O(sqrt(n))
2 // 區間為 [l, r]
3 for(int i=1; i<=n; i++){
4     int l = i, r = n/(n/i);
5     i = r;
6     ans.push_back(r);
7 }
```

7.20 最大質因數 [ca5e52]

```
1 void max_fac(int n, int &ret){
2     if (n<=ret || n<2) return;
3     if (isprime(n)){
4         ret = max(ret, n);
5         return;
6     }
7
8     int p = Pollard_Rho(n);
9     max_fac(p, ret), max_fac(n/p, ret);
10 }
```

7.21 歐拉公式 [7e25f3]

```
1 // phi(n) = 小於 n 並與 n 互質的正整數數量。
2 // O(sqrt(n)) · 回傳 phi(n)
3 int phi(int n){
4     int ret = n;
5
6     for (int i=2; i*i<=n; i++){
7         if (n%i==0){
8             while (n%i==0) n /= i;
9             ret = ret*(i-1)/i;
10        }
11    }
12    if (n>1) ret = ret*(n-1)/n;
13
14    return ret;
15 }
16
17 // O(n) · 回傳 1~n 的 phi 值
18 vector<int> phi_1_to_n(int n) {
19     vector<int> primes, lpf(n+1), phi(n+1);
20     phi[0] = 0; phi[1] = 1;
21     for (int i = 2; i <= n; ++i) {
22         if (lpf[i] == 0) {
23             primes.push_back(i);
24             lpf[i] = i;
25         }
26         for (auto p : primes) {
27             if (i * p > n) break;
```

```

28         lpf[i * p] = p;
29         if (i % p == 0) break;
30     }
31     bool b = i % (lpf[i] * lpf[i]);
32     phi[i] = phi[i / lpf[i]] * (lpf[i] - b);
33 }
34 return phi;
35 }

```

7.22 歐拉定理

$$a^n \equiv \begin{cases} a^{n \bmod \varphi(m)}, & \gcd(a, m) = 1, \\ a^n, & \gcd(a, m) \neq 1, n < \varphi(m), \\ a^{(n \bmod \varphi(m)) + \varphi(m)}, & \gcd(a, m) \neq 1, n \geq \varphi(m). \end{cases} \quad (\bmod m)$$

7.23 錯排公式

錯排公式：(n 個人中，每個人皆不再原來位置的組合數)

$$dp_i = \begin{cases} 1 & i = 0 \\ 0 & i = 1 \\ (i-1)(dp_{i-1} + dp_{i-2}) & \text{otherwise} \end{cases}$$

8 String

8.1 AC automation [018290]

```

1 struct ACAutomation{
2     vector<vector<int>> go;
3     vector<int> fail, match, pos;
4     int sz = 0; // 有效節點為 [0, sz] · 開陣列的時候要小心 !!!
5
6     ACAutomation(int n) : go(n, vector<int>(26)), fail(n), match(n) {}
7
8     void add(string s){
9         int now = 0;
10        for (char c : s){
11            if (!go[now][c-'a']) go[now][c-'a'] = ++sz;
12            now = go[now][c-'a'];
13        }
14        pos.push_back(now);
15    }
16
17    void build(){
18        queue<int> que;
19        for (int i=0 ; i<26 ; i++){
20            if (go[0][i]) que.push(go[0][i]);
21        }
22        while (que.size()){
23            int u = que.front();
24            que.pop();
25            for (int i=0 ; i<26 ; i++){
26                if (go[u][i]){
27                    fail[go[u][i]] = go[fail[u]][i];
28                    que.push(go[u][i]);
29                }else go[u][i] = go[fail[u]][i];
30            }
31        }
32    }
33
34    // counting pattern

```

```

35 void buildMatch(string &s){
36     int now = 0;
37     for (char c : s){
38         now = go[now][c-'a'];
39         match[now]++;
40     }
41
42     vector<int> in(sz+1), que;
43     for (int i=1 ; i<=sz ; i++) in[fail[i]]++;
44     for (int i=1 ; i<=sz ; i++) if (in[i]==0) que.push_back(i);
45
46     for (int i=0 ; i<que.size() ; i++){
47         int now = que[i];
48         match[fail[now]] += match[now];
49         if (--in[fail[now]]==0) que.push_back(fail[now]);
50     }
51 };

```

8.2 Enumerate Runs [74311c]

```

1 // returns [minimum_period, ans_l, ans_r] (開區間, 0-based)
2 // (ans_r - ans_l + 1) >= minimum_period * 2
3 vector<array<int, 3>> enumerate_run(string s) {
4     int n = s.size();
5     if (n <= 1) return {};
6     vector<vector<pair<int, int>>> stor(n+1);
7
8     auto sub = [&](string l, string r) {
9         int a = l.size(), b = r.size();
10        auto zl = z_function(string(l.rbegin(), l.rend()));
11        auto zr = z_function(r + l + r); // Z_Algorithm.cpp
12        vector<array<int, 3>> res;
13        for (int p = 1; p <= a; p++) {
14            int x = (p == a) ? p : min(zl[p] + p, a);
15            int y = min(zr[a + b - p], b);
16            if (x + y >= 2 * p) res.push_back({p, x, y});
17        }
18        return res;
19    };
20
21    auto dfs = [&](auto self, int l, int r) -> void {
22        if (r - l <= 1) return;
23        int m = (l + r) >> 1;
24        self(self, l, m); self(self, m, r);
25        auto sl = s.substr(l, m-l), sr = s.substr(m, r-m);
26        for (auto [p, a, b] : sub(sl, sr))
27            stor[p].push_back({m - a, m + b});
28        reverse(sl.begin(), sl.end());
29        reverse(sr.begin(), sr.end());
30        for (auto [p, a, b] : sub(sr, sl))
31            stor[p].push_back({m - b, m + a});
32    };
33    dfs(dfs, 0, n);
34
35    vector<array<int, 3>> runs;
36    set<pair<int, int>> done;
37    for (int p = 1; p <= n; p++) {
38        auto& v = stor[p];
39        sort(v.begin(), v.end(), [](auto& x, auto& y) {
40            return (x.first != y.first) ? (x < y)
41                : (x.second > y.second);
42        });
43        int mx = -1;
44        for (auto &[l, r] : v) if (r > mx) {

```

```

45         mx = r;
46         if (done.insert({l, r}).second)
47             runs.push_back({p, l, r - 1});
48     }
49 }
50 sort(runs.begin(), runs.end());
51 return runs;
52 }

```

8.3 Hash [942f42]

```

1 mt19937 seed(chrono::steady_clock::now().time_since_epoch().
2   count());
3 int rng(int l, int r){
4     return uniform_int_distribution<int>(l, r)(seed);
5 }
6 int A = rng(1e5, 8e8);
7 const int B = 1e9+7;
8
9 // 2f6192
10 struct RollingHash{
11     vector<int> Pow, Pre;
12     RollingHash(string s = ""){
13         Pow.resize(s.size());
14         Pre.resize(s.size());
15
16         for (int i=0 ; i<s.size() ; i++){
17             if (i==0){
18                 Pow[i] = 1;
19                 Pre[i] = s[i];
20             }else{
21                 Pow[i] = Pow[i-1]*AB;
22                 Pre[i] = (Pre[i-1]*A+s[i])%B;
23             }
24         }
25         return;
26     }
27
28     int get(int l, int r){ // 取得 [l, r] 的數值
29         if (l==0) return Pre[r];
30         int res = (Pre[r]-Pre[l-1]*Pow[r-l+1])%B;
31         if (res<0) res += B;
32         return res;
33     }
34 };

```

8.4 KMP [287490]

```

1 // KMP[i] = s[0...i] 的最長共同前後綴長度 · KMP[KMP[i]-1] 可以跳 fail link
2 vector<int> KMP(string &s){
3     vector<int> ret(s.size());
4     for (int i=1 ; i<s.size() ; i++){
5         int j = ret[i-1];
6         while (j && s[i]!=s[j]) j = ret[j-1];
7         ret[i] = j + (s[i]==s[j]);
8     }
9     return ret;
10 }

```

8.5 Manacher [199269]

```

1 // 回傳所有最長 (無法往兩側擴展) 迴文的區間 [l, r], 0-based
2 vector<array<int, 2>> Manacher(string str) {
3     string tmp = "$#";
4     for(char i : str) {
5         tmp += i;
6         tmp += '#';
7     }
8
9     int m = tmp.size(), mx = 0, id = 0;
10    vector<int> p(m);
11    vector<array<int, 2>> res;
12    for(int i=1 ; i<m ; i++) {
13        p[i] = mx > i ? min(p[id*2-i], mx-i) : 1;
14        while(tmp[i+p[i]] == tmp[i-p[i]]) p[i]++;
15        if (mx < i + p[i]) mx = i + p[i], id = i;
16
17        if (p[i] > 1) {
18            int r = (i / 2) + (p[i] / 2) - 2 + (i & 1);
19            res.push_back({r - p[i] + 2, r});
20        }
21    }
22    return res;
23 }

```

8.6 Min Rotation [b24786]

```

1 int minRotation(string s) {
2     int a = 0, n = s.size();
3     s += s;
4
5     for (int b=0 ; b<n ; b++){
6         for (int k=0 ; k<n ; k++){
7             if (a+k==b || s[a+k]<s[b+k]){
8                 b += max(0LL, k-1);
9                 break;
10            }
11            if (s[a+k]>s[b+k]){
12                a = b;
13                break;
14            }
15        }
16    }
17    return a;
18 }

```

8.7 Suffix Array [47a062]

```

1 // 注意 · 當 |s|=1 時 · lcp 不會有值 · 務必測試 |s|=1 的 case
2 struct SuffixArray {
3     string s;
4     vector<int> sa, lcp;
5
6     // 0eea96
7     template<typename T>
8     void sais(const T s, int n, int *sa, int *lar, int *p,
9              int *t, int A) {
10        int *rnk = p + n, *q = p + n / 2, *bkt = lar + A;
11        int m = 0, i, j, x = t[n - 1] = 1, y = rnk[0] = -1,
12            cnt = -1;
13
14        for (i=n-2 ; ~i ; i--) t[i] = (s[i] == s[i + 1] ? t[i
15            + 1] : s[i] < s[i + 1]);
16    }

```

```

13    for (i=1 ; i<n ; i++) rnk[i] = t[i] && !t[i-1] ? (p[m
14        ]=i, m++) : -1;
15    fill_n(lar, A, 0);
16    for (i=0 ; i<n ; i++) ++lar[s[i]];
17    for (i=1 ; i<A ; i++) lar[i] += lar[i-1];
18    auto pushS = [&](int x) { sa[--bkt[s[x]]] = x; };
19    auto induce_sort = [&](int* v) {
20        fill_n(sa, n, 0);
21        copy_n(lar, A, bkt);
22        for (i=m-1 ; ~i ; i--) pushS(v[i]);
23        copy_n(lar, A - 1, bkt + 1);
24        for (i=0 ; i<n ; i++) if (sa[i] && !t[sa[i] - 1])
25            pushL(sa[i] - 1);
26        copy_n(lar, A, bkt);
27        for (i=n-1 ; ~i ; i--) if (sa[i] && t[sa[i] - 1])
28            pushS(sa[i] - 1);
29    };
30    induce_sort(p);
31    for (i=0 ; i<n ; i++){
32        if (~x=rnk[sa[i]]){
33            j = y < 0 || memcmp(s + p[x], s + p[y], (p[x
34                + 1] - p[x]) * sizeof(s[0]));
35            q[y = x] = cnt += j;
36        }
37    }
38    if (cnt+1<m) sais(q, m, sa, bkt, rnk, t + n, cnt + 1)
39        ;
40    else for (i=0 ; i<m ; i++) sa[q[i]] = i;
41    for (i=0 ; i<m ; i++) q[i] = p[sa[i]];
42    induce_sort(q);
43 }
44
45 // da9ddf, O(n + lim), lim = max{s[i]}
46 template<typename T>
47 SuffixArray(const T& _s) : s(_s.begin(), _s.end()) {
48     s.push_back(-1);
49     for (auto &i:s) i += 1;
50     int n = s.size(), lim = *max_element(s.begin(), s.end
51         ()) + 5;
52     sa.resize(n);
53     lcp.resize(n);
54     vector<int> bkt(n + lim * 2), p(n * 2), t(n * 2),
55         rank(n);
56     sais(&s[0], n, &sa[0], &bkt[0], &p[0], &t[0], lim);
57
58     for (int i=1 ; i<n ; i++) rank[sa[i]] = i;
59     for (int i=0, j, k=0 ; i<n-1 ; lcp[rank[i++]]=k){
60         for (k && k--, j=sa[rank[i]-1] ; i+k<s.size() &&
61             j+k<s.size() && s[i+k]==s[j+k] ; k++);
62     }
63
64     sa.erase(sa.begin());
65     lcp.erase(lcp.begin(), lcp.begin()+2);
66     s.pop_back();
67 }
68
69 // f49583
70 vector<int> pos; // pos[i] = i 這個值在 pos 的哪個地方
71 SparseTable st;
72 void init_lcp(){
73     pos.resize(sa.size());
74     for (int i=0 ; i<sa.size() ; i++){
75         pos[sa[i]] = i;
76     }
77 }

```

```

70     if (lcp.size()){
71         st.build(lcp);
72     }
73 }
74
75 // 用之前記得 init
76 // 查詢「sa 上的位置」的 x 跟 y 的 lcp
77 int get_lcp(int x, int y){
78     if (x==y) return s.size()-x;
79     if (x>y) swap(x, y);
80     return st.query(x, y);
81 }
82
83 // 回傳 [l1, r1] 跟 [l2, r2] 的 lcp · 0-based
84 int get_lcp(int l1, int r1, int l2, int r2){
85     int pos_1 = pos[l1], len_1 = r1-l1+1;
86     int pos_2 = pos[l2], len_2 = r2-l2+1;
87     if (pos_1>pos_2){
88         swap(pos_1, pos_2);
89         swap(len_1, len_2);
90     }
91
92     if (l1==l2){
93         return min(len_1, len_2);
94     }else{
95         return min({st.query(pos_1, pos_2), len_1, len_2
96             });
97     }
98 }
99
100 // 檢查 [l1, r1] 跟 [l2, r2] 的大小關係 · 0-based
101 // 如果前者小於後者 · 就回傳 <0 · 相等就回傳 =0 · 否則回傳
102 // >0
103 // 5b8db0
104 int substring_cmp(int l1, int r1, int l2, int r2){
105     int len_1 = r1-l1+1;
106     int len_2 = r2-l2+1;
107     int res = get_lcp(l1, r1, l2, r2);
108
109     if (res<len_1 && res<len_2){
110         return s[l1+res]-s[l2+res];
111     }else if (len_1==res && len_2==res){
112         return 0;
113     }else{
114         return len_1==res ? -1 : 1;
115     }
116 }
117
118 // 對於位置在 <=p 的後綴 · 找離他左邊/右邊最接近位置 >p 的
119 // 後綴的 lcp · 0-based
120 // pre[i] = s[i] 離他左邊最接近位置 >p 的後綴的 lcp · 0-
121 // based
122 // suf[i] = s[i] 離他右邊最接近位置 >p 的後綴的 lcp · 0-
123 // based
124 // da12fa
125 pair<vector<int>, vector<int>> get_left_and_right_lcp(int
126     p){
127     vector<int> pre(p+1);
128     vector<int> suf(p+1);
129
130     { // build pre
131         int now = 0;
132         for (int i=0 ; i<s.size() ; i++){
133             if (sa[i]<=p){

```

```

128         pre[sa[i]] = now;
129         if (i < lcp.size()) now = min(now, lcp[i]);
130     }else{
131         if (i < lcp.size()) now = lcp[i];
132     }
133 }
134 { // build suf
135     int now = 0;
136     for (int i=s.size()-1 ; i>=0 ; i--){
137         if (sa[i]<=p){
138             suf[sa[i]] = now;
139             if (i-1>=0) now = min(now, lcp[i-1]);
140         }else{
141             if (i-1>=0) now = lcp[i-1];
142         }
143     }
144 }
145 }
146
147 return {pre, suf};
148 }
149 };

```

8.8 Suffix Automata [efe54a]

```

1 // root node is 0
2 // node -> strings with the same endpos set,
3 //     length in range [len[fa] + 1, len]
4 // link -> longest suffix with different endpos set
5 // len -> longest path length from root
6 // cnt -> size of endpos set
7 // node's endpos set -> pos (1-based) in the subtree (link)
8 //     of node (Some node have pos = 0 and should be ignored)
9 struct SAM {
10     static constexpr int N = (int)(1e5 + 10), Z = 26;
11     int ch[2 * N][Z], len[2 * N], link[2 * N], pos[2 * N], cnt
12         [2 * N], sz;
13     int newnode() {
14         fill_n(ch[sz], Z, 0);
15         len[sz] = link[sz] = pos[sz] = cnt[sz] = 0;
16         return sz++;
17     }
18     void build(string s) {
19         sz = 0, newnode(), link[0] = -1;
20         int lst = 0;
21         for (int i = 0; i < s.size(); ++i) {
22             int c = s[i] - 'a'; // you may need to change to 'A'
23             int cur = newnode();
24             len[cur] = len[lst] + 1, pos[cur] = i + 1;
25             int p = lst;
26             while (~p && !ch[p][c])
27                 ch[p][c] = cur, p = link[p];
28             if (p == -1) link[cur] = 0;
29             else {
30                 int q = ch[p][c];
31                 if (len[p] + 1 == len[q]) {
32                     link[cur] = q;
33                 } else {
34                     int nxt = newnode();
35                     len[nxt] = len[p] + 1, link[nxt] = link[q];
36                     pos[nxt] = 0;
37                     for (int j = 0; j < Z; ++j)
38                         ch[nxt][j] = ch[q][j];
39                     while (~p && ch[p][c] == q)
40                         ch[p][c] = nxt, p = link[p];

```

```

39         link[q] = link[cur] = nxt;
40     }
41 }
42 cnt[cur]++, lst = cur;
43 }
44 vector<int> p(sz);
45 iota(p.begin(), p.end(), 0);
46 sort(p.begin(), p.end(), [&](int i, int j) {
47     return len[i] > len[j];
48 });
49 for (int i = 0; i < sz; ++i) if (~link[p[i]])
50     cnt[link[p[i]]] += cnt[p[i]];
51 }
52 };

```

8.9 Wildcard Matching [e93074]

```

1 const int B = 27;
2 string p;
3 for (int i=0 ; i<B ; i++) p += ('a'+i);
4 p += '*';
5 vector<vector<double>> res(B);
6 for (int i=0 ; i<B ; i++){
7     vector<double> ss, tt;
8     for (auto x : s) ss.push_back(x==p[i] || x=='*');
9     for (auto x : t) tt.push_back(x==p[i] || x=='*');
10    reverse(tt.begin(), tt.end());
11    res[i] = PolyMul(ss, tt);
12 }
13 for (int i=t.size()-1 ; i+t.size()-1<res[0].size() ; i++){
14     int total = 0;
15     for (int j=0 ; j<B-1 ; j++) total += (int)abs(round(res[j
16         ][i]));
17     total -= 25*(int)abs(round(res[26][i]));
18     cout << (total==t.size());

```

8.10 Z Algorithm [9d559a]

```

1 // z[i] 回傳 s[0...] 跟 s[i...] 的 lcp, z[0] = 0
2 vector<int> z_function(string s){
3     vector<int> z(s.size());
4     int l = -1, r = -1;
5     for (int i=1 ; i<s.size() ; i++){
6         z[i] = i>=r ? 0 : min(r-i, z[i-1]);
7         while (i+z[i]<s.size() && s[i+z[i]]==s[z[i]]) z[i]++;
8         if (i+z[i]>r) l=i, r=i+z[i];
9     }
10    return z;
11 }

```